

Estimating hybrid choice models with Biogeme

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The package Biogeme (`biogeme.epfl.ch`) is designed to estimate the parameters of various models using maximum likelihood estimation. It is particularly designed for discrete choice models. In this document, we present how to estimate choice models involving latent variables: hybrid choice models.

We assume that the reader is already familiar with discrete choice models, and has successfully installed Biogeme. This document has been written using Biogeme 3.3.3.

1 Models and notations

The literature on discrete choice models with latent variables is vast (Walker, 2001, Ashok et al., 2002, Greene and Hensher, 2003, Ben-Akiva, Walker, Bernardino, Gopinath, Morikawa and Polydoropoulou, 2002, Ben-Akiva et al., 2002, to cite just a few early references). This document presents a practical introduction to the specification and estimation of hybrid choice models using Biogeme. It introduces the main concepts and notation underlying these models, discusses the associated normalization issues, and illustrates their implementation through a sequence of progressively richer examples. The examples show how latent variables, structural equations, measurement equations, and choice models can be combined within a unified estimation framework.

1.1 Structural equations

A *latent variable* is a variable that cannot be directly observed. It is typically modeled using a **structural equation**, which expresses the latent variable as a function of observed (explanatory) variables and an error term. An asterisk (*) is used to denote latent variables, distinguishing them from observed variables. A general form of such a structural equation is:

$$\mathbf{x}_{nk}^* = \mathbf{x}^*(\mathbf{x}_n; \boldsymbol{\psi}_k) + \boldsymbol{\omega}_{nk}, \quad (1)$$

where $\mathbf{n}$ indexes individuals, $\mathbf{x}_{nk}^*$ is the k th latent variable of interest, $\mathbf{x}_n$ is a vector of observed explanatory variables, $\boldsymbol{\psi}_k$ is a vector of parameters to be estimated, and $\boldsymbol{\omega}_{nk}$ is a stochastic error term.

A common specification assumes a linear functional form with i.i.d. normally distributed error terms:

$$\mathbf{x}_{nk}^* = \sum_s \boldsymbol{\psi}_{sk} \mathbf{x}_{ns} + \sigma_{\omega k} \boldsymbol{\omega}_{nk}, \quad (2)$$

where $\boldsymbol{\omega}_{nk} \sim \mathcal{N}(0, 1)$, and $\sigma_{\omega k}$ is a scaling parameter for the error term.

Information about latent variables is obtained indirectly through *measurements*, which are observable manifestations of the underlying latent constructs. For example, in discrete choice models, utility is not directly observed but is inferred from the choices individuals make. The relationship between a latent variable and its associated measurements is described by **measurement equations**. The specific form of these equations depends on the nature of the observed measurements (e.g., continuous, or ordinal).

1.2 Measurement equations: the continuous case

Since latent variables cannot be directly observed, analysts rely on indirect measurements to infer their values. A common approach involves asking respondents to rate the perceived magnitude of the latent construct on an arbitrary scale. For example: “How would you rate the level of pain that you are experiencing, from 0 (no pain) to 10 (worst pain imaginable)?”

Each such rating is referred to as an *indicator*, indexed by $\ell = 1, \dots, L_n$, and is modeled using a **measurement equation**. This equation relates the observed indicator to the latent variables and, sometimes, other explanatory variables:

$$I_{n\ell} = I_\ell(\mathbf{x}_n, \mathbf{x}_n^*; \boldsymbol{\lambda}_\ell) + \mathbf{v}_{n\ell}, \quad \forall \ell = 1, \dots, L_n, \forall \mathbf{n}, \quad (3)$$

where $I_{n\ell}$ denotes the response provided by individual $\mathbf{n}$ for indicator ℓ , $\mathbf{x}_n^*$ is the latent variable of interest (e.g., pain perception), $\mathbf{x}_n$ is a vector of observed explanatory variables (such as socio-demographic characteristics), λ_ℓ is a vector of parameters to be estimated, and $\mathbf{v}_{n\ell}$ is the random error term.

A common specification of the measurement function assumes linearity and i.i.d. normally distributed errors:

$$I_{n\ell} = \lambda_{\ell 0} + \sum_k \lambda_{\ell k} \mathbf{x}_{nk}^* + \sigma_{v\ell} \mathbf{v}_{n\ell}, \quad \forall \ell, \quad (4)$$

where $\lambda_{\ell k}$ are unknown parameters to be estimated, $\sigma_{v\ell}$ is an indicator-specific scale parameter, and $\mathbf{v}_{n\ell} \sim \mathbf{N}(0, 1)$.

If we observe a vector of continuous indicators $\mathbf{I}_n = (I_{n1}, \dots, I_{nL_n})$ for individual $\mathbf{n}$, the contribution to the likelihood function, *conditional on the latent variables* $\mathbf{x}_n^*$, and assuming conditional independence of the indicators given $\mathbf{x}_n^*$, is given by:

$$\prod_{\ell=1}^{L_n} \frac{1}{\sigma_{v\ell}} \phi \left(\frac{I_{n\ell} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} \mathbf{x}_{nk}^*}{\sigma_{v\ell}} \right), \quad (5)$$

where $\phi(\cdot)$ denotes the probability density function (pdf) of the standard normal distribution.

If other types of observations are available for the same individual (such as discrete choices), the corresponding components of the likelihood can be multiplied with the expression above. Once all relevant components are combined, the latent variables must be integrated out, as discussed later.

For instance, if the continuous indicators are the only data available for individual $\mathbf{n}$, the contribution to the unconditional likelihood becomes:

$$\int_{\mathbf{x}_n^*} \left[\prod_{\ell=1}^{L_n} \frac{1}{\sigma_{v\ell}} \phi \left(\frac{I_{n\ell} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} \mathbf{x}_{nk}^*}{\sigma_{v\ell}} \right) \right] f(\mathbf{x}_n^* | \mathbf{x}_n; \boldsymbol{\alpha}) d\mathbf{x}_n^*. \quad (6)$$

where $f(\mathbf{x}_n^* | \mathbf{x}_n; \boldsymbol{\alpha})$ is the conditional joint probability density function of the latent variables $\mathbf{x}_n^*$ given the observed explanatory variables $\mathbf{x}_n$. The vector $\boldsymbol{\alpha}$ contains all parameters appearing in the structural equations (2), including the coefficients $\boldsymbol{\psi}$ and the scale parameters of the structural disturbances. As this integral does not have a closed-form expression, it is approximated using Monte Carlo integration (see Bierlaire, 2019 for a discussion about performing Monte-Carlo integration with Biogeme).

1.3 Measurement equation: the ordinal case

Another type of indicator arises when respondents are asked to evaluate a statement using an ordinal scale. A typical context for this type of measurement is the use of a Likert scale (Likert, 1932), where individuals express their degree of agreement or disagreement with a given statement. For example:

“I believe that my own actions have an impact on the planet.”

Response options: strongly agree (2), agree (1), neutral (0), disagree (−1), strongly disagree (−2).

In practice, ordinal indicators are sometimes treated as if they were continuous variables. This approximation is common when the scale contains several ordered categories and the distances between adjacent categories are assumed to be approximately equal. In that case, the indicator can be modeled using the continuous measurement equation presented in the previous section. If, however, the discrete and ordered nature of the indicator is to be explicitly taken into account, a different specification is required. The approach presented below models the indicator as an ordered discrete variable and directly accounts for the ordinal structure of the response scale.

To model these types of indicators, we represent the observed measurement as an *ordered discrete variable* $I_{n\ell}$, which takes values in a finite, ordered set $\{j_1, j_2, \dots, j_{M_\ell}\}$. The measurement equation involves two stages:

Step 1: Latent response formulation. We first define a continuous response variable, as explained in Section 1.2, except that it happens to be unobserved (latent) in this case:

$$I_{n\ell}^* = I_\ell^*(\mathbf{x}_n, \mathbf{x}_n^*; \lambda_\ell) + \mathbf{v}_{n\ell}, \quad (7)$$

where $I_{n\ell}^*$ is a continuous latent variable underlying the reported response, $\mathbf{x}_n^*$ is a vector of relevant latent variables (e.g., environmental concern), $\mathbf{x}_n$ is a vector of observed explanatory variables (e.g., age, income), λ is a vector of parameters to be estimated, and $\mathbf{v}_{n\ell}$ is a random error term.

Step 2: Discretization via thresholds. Since $I_{n\ell}^*$ is not observed, we relate it to the reported discrete measurement $I_{n\ell}$ through a set of threshold parameters:

$$I_{n\ell} = \begin{cases} j_1 & \text{if } I_{n\ell}^* < \tau_{\ell 1}, \\ j_2 & \text{if } \tau_{\ell 1} \leq I_{n\ell}^* < \tau_{\ell 2}, \\ \vdots & \\ j_m & \text{if } \tau_{\ell, m-1} \leq I_{n\ell}^* < \tau_{\ell m}, \\ \vdots & \\ j_M & \text{if } \tau_{\ell, M_\ell-1} \leq I_{n\ell}^*, \end{cases} \quad (8)$$

where $\tau_{\ell 1}, \dots, \tau_{\ell, M_\ell-1}$ are threshold parameters to be estimated, satisfying the ordering constraint:

$$\tau_{\ell 1} \leq \tau_{\ell 2} \leq \dots \leq \tau_{\ell, M_\ell-1}. \quad (9)$$

The formulation above associates a distinct set of threshold parameters $\tau_{\ell 1}, \dots, \tau_{\ell, M_\ell-1}$ with each indicator ℓ . In practice, this specification may involve a large number of parameters when many ordinal indicators are included in the model. A common simplification is to impose a common set of thresholds for groups of indicators that share the same response scale and are intended to measure similar constructs. In that case, the thresholds are no longer indicator-specific and the index ℓ can be dropped.

From a statistical perspective, this restriction reduces the flexibility of the model, as the relative distances between response categories are assumed to be identical across the corresponding indicators. The model is therefore more constrained than the fully general specification. However, the resulting reduction in the number of parameters can be substantial, leading to more stable estimation, shorter computation times, and improved interpretability. Whether such a simplification is appropriate depends on the application and on the degree of similarity among the indicators.

The case study presented in Section 4 adopts this simplified specification and uses a common set of thresholds across all ordinal indicators.

Defining $\tau_{\ell 0} = -\infty$ and $\tau_{\ell, M_\ell} = +\infty$, it simplifies to

$$I_{n\ell} = j_m \text{ if } \tau_{\ell, m-1} \leq I_{n\ell}^* < \tau_{\ell m}, \quad m = 1, \dots, M_\ell. \quad (10)$$

If we consider a linear specification,

$$I_{n\ell}^* = \lambda_{\ell 0} + \sum_k \lambda_{\ell k} \mathbf{x}_{nk}^* + \sigma_{v\ell} \mathbf{v}_{n\ell}, \quad \forall \ell, \quad (11)$$

where the error term $\mathbf{v}_{n\ell} \sim N(0, 1)$, the probability of observing category j_m is equal to the probability that the latent response $I_{n\ell}^*$ falls between the two thresholds defining that category, namely $\tau_{\ell, m-1}$ and $\tau_{\ell m}$. Therefore, the contribution of indicator ℓ for observation $\mathbf{n}$ to

the likelihood function, conditional on the latent variables, can be derived directly from the distribution of the disturbance term:

$$\begin{aligned}
\Pr(I_{n\ell} = j_m \mid \mathbf{x}_n^*, \mathbf{x}_n; \lambda_\ell, \sigma_{v\ell}) &= \Pr(\tau_{\ell, m-1} \leq I_{n\ell}^* < \tau_{\ell, m}) \\
&= \Pr(I_{n\ell}^* < \tau_{\ell, m}) - \Pr(I_{n\ell}^* < \tau_{\ell, m-1}) \\
&= \Pr\left(v_{n\ell} \leq \frac{\tau_{\ell, m} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} x_{nk}^*}{\sigma_{v\ell}}\right) \\
&\quad - \Pr\left(v_{n\ell} \leq \frac{\tau_{\ell, m-1} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} x_{nk}^*}{\sigma_{v\ell}}\right) \quad (12) \\
&= \Phi\left(\frac{\tau_{\ell, m} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} x_{nk}^*}{\sigma_{v\ell}}\right) \\
&\quad - \Phi\left(\frac{\tau_{\ell, m-1} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} x_{nk}^*}{\sigma_{v\ell}}\right),
\end{aligned}$$

where j_m is the observed category for respondent n and indicator ℓ , and $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

The probability of observing a category is therefore obtained as the difference between two cumulative probabilities. Intuitively, the latent response must fall within the interval delimited by the two thresholds associated with that category. The above specification computes this probability using the normal distribution of the disturbance term. It is known as the *ordered probit model* and is widely used for modeling ordinal responses that depend on latent constructs.

An alternative specification is obtained by assuming that the disturbance term follows a standard logistic distribution,

$$v_{n\ell} \sim \text{Logistic}(0, 1),$$

instead of a standard normal distribution. In that case, the cumulative distribution function $\Phi(\cdot)$ in Eq. (12) is replaced by the logistic cumulative distribution function

$$F(t) = \frac{1}{1 + \exp(-t)},$$

yielding the *ordered logit model*. The resulting category probabilities are therefore

$$\Pr(I_{n\ell} = j_m \mid \mathbf{x}_n^*, \mathbf{x}_n) = F\left(\frac{\tau_{\ell, m} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} x_{nk}^*}{\sigma_{v\ell}}\right) - F\left(\frac{\tau_{\ell, m-1} - \lambda_{\ell 0} - \sum_k \lambda_{\ell k} x_{nk}^*}{\sigma_{v\ell}}\right). \quad (13)$$

The ordered probit and ordered logit models differ only through the assumed distribution of the disturbance term. Since the normal and logistic cumulative distribution functions have similar shapes, the two specifications often lead to very similar empirical results. Both models are implemented in Biogeme and illustrated in the examples of Section 4.

If we observe a vector of ordinal indicators $\mathbf{I}_n = (I_{n1}, \dots, I_{nL_n})$ for individual n , the contribution to the likelihood function, *conditional on the latent variables* $\mathbf{x}_n^*$, is given by:

$$\prod_{\ell=1}^{L_n} \Pr(I_{n\ell} = j_m \mid \mathbf{x}_n^*, \mathbf{x}_n; \lambda_\ell, \Sigma_{v\ell}). \quad (14)$$

As in the continuous case, if other types of observations are available for the same individual (such as choices), the corresponding components of the likelihood can be multiplied with the expression above. Once all relevant components are combined, the latent variables must be integrated out, as discussed later.

If the ordinal indicators are the only data available for individual n , the contribution to the unconditional likelihood becomes:

$$\int_{\mathbf{x}_n^*} \left[\prod_{\ell=1}^{L_n} \Pr(I_{n\ell} = j_m \mid \mathbf{x}_n^*, \mathbf{x}_n; \lambda_\ell, \Sigma_{v\ell}) \right] f(\mathbf{x}_n^* \mid \mathbf{x}_n; \alpha) d\mathbf{x}_n^*, \quad (15)$$

where $f(\mathbf{x}_n^*|\mathbf{x}_n; \boldsymbol{\alpha})$ is the conditional pdf of the latent variables given the observed explanatory variables, and $\boldsymbol{\alpha}$ denotes the parameters of the structural equations. It includes the coefficients $\boldsymbol{\psi}$ of the structural equations as well as the parameters governing the distribution of the structural disturbances in Equation (2). Again, this integral is approximated using Monte-Carlo integration.

2 Normalization and identification

Models with latent variables contain parameters that are not uniquely identified from the data unless additional restrictions are imposed. These restrictions, referred to as *normalizations*, are not behavioral assumptions; rather, they fix the arbitrary units of measurement (location and scale) that are inherent to latent constructs and ordinal measurement schemes.

This section explains why normalization is required, where non-identification arises, and how to impose a consistent and non-redundant set of normalizations. We distinguish between normalizations that identify the *latent variables themselves* and those that identify the *ordinal measurement layers*. In principle, there are infinitely many equivalent normalization schemes, all leading to statistically equivalent models and identical behavioral conclusions. The choice of a normalization is therefore a matter of convention rather than substance. Among the possible approaches, we present the reference-indicator strategy, which is widely used because of its interpretability.

2.1 Why normalization is required

Latent-variable models are invariant to specific transformations of latent variables and associated parameters. Without normalization, multiple parameter vectors generate exactly the same likelihood. In such cases, estimation is ill-posed: the likelihood exhibits flat directions, standard errors are unreliable, and numerical optimization or Bayesian sampling may be unstable.

At the level of latent variables, two fundamental sources of non-identification arise:

- **Location invariance:** the origin of a latent variable is arbitrary;
- **Scale invariance:** the unit in which a latent variable is measured is arbitrary.

Both must be addressed for each latent variable in the model.

The need for normalization is closely analogous to identification issues in logit and related random utility models. Utilities are latent and only differences in utility matter; as a consequence, utilities are invariant to the addition of a constant and to multiplication by a positive scalar. Standard practice therefore requires fixing one alternative-specific constant (location) and fixing the utility scale, typically by normalizing the variance of the error term.

Latent variables play a similar role: they enter the likelihood only through systematic components and relative comparisons. Measurement equations — whether continuous or ordinal — therefore require explicit normalizations to fix both location and scale.

2.2 Normalization of latent variables

Consider the linear structural and measurement equations introduced in Section 1. If measurement intercepts are estimated, any shift of a latent variable can be absorbed by a compensating shift in the intercepts. Likewise, any rescaling of a latent variable can be offset by rescaling factor loadings and structural parameters. As a result, neither the location nor the scale of the latent variable is identified from the data alone.

For each latent variable, exactly two normalizations are therefore required:

- one to fix its *location*;

- one to fix its *scale*.

These two normalizations are necessary and sufficient to remove the two degrees of freedom associated with location and scale invariance. Any additional restriction affecting these dimensions would be redundant and may lead to over-constrained specifications.

A practical and interpretable approach is the *reference-indicator strategy*. Consider the measurement equation (4). For each latent variable k , one indicator $\ell(k)$ is selected as its reference indicator. Location is fixed by setting the corresponding intercept parameter $\lambda_{\ell(k)0}$ to zero, and scale is fixed by setting the corresponding factor loading $\lambda_{\ell(k)k}$ to either $+1$ or -1 . The sign choice determines the orientation of the latent scale and should be chosen to preserve semantic consistency between the latent variable and its indicators.

Once these two constraints are imposed, remaining parameters—such as measurement error variances and non-reference loadings—become meaningful and interpretable.

2.3 Normalization of ordinal measurement models

We now consider ordinal indicators modeled using ordered response specifications, such as the ordered probit and ordered logit models introduced in Section 1.3. In this case, the observed response reveals only the interval in which an underlying latent response lies. This introduces additional sources of non-identification beyond those associated with the latent variables themselves.

For each ordered probit measurement equation, two additional invariances arise:

- **Threshold-location invariance:** adding the same constant to the latent response, the measurement intercept, and all thresholds leaves observed probabilities unchanged;
- **Threshold-scale invariance:** multiplying the latent response, all thresholds, and the measurement error standard deviation by the same positive constant leaves probabilities unchanged.

These invariances are *specific to each independent set of thresholds*, that is, to each group of indicators that shares a common latent response scale and threshold parameters. Consequently, normalization of the ordinal measurement layer must be performed *separately for each threshold system*, even when multiple indicators share the same response scale.

Identification therefore requires, for each threshold system, one normalization to fix *location* and one to fix *scale*.

The phrase *threshold system* is important here. If each ordinal indicator has its own thresholds, then each indicator defines its own threshold system and requires its own location and scale normalizations. If, instead, several indicators are constrained to share the same thresholds, these indicators form a single threshold system. In that case, the location and scale normalizations are imposed only once for the group, not once per indicator.

For example, if a set of Likert indicators is modeled with common thresholds

$$\tau_1, \dots, \tau_{M-1},$$

then one location normalization, such as fixing one threshold or imposing symmetry around zero, and one scale normalization, such as fixing the ordered-probit error variance, identify the ordinal measurement layer for the whole group. Imposing the same normalizations separately for each indicator in the group would be redundant, because the indicators no longer have separate threshold locations or scales.

Using common thresholds therefore reduces not only the number of threshold parameters, but also the number of ordinal-layer normalizations required. The general rule is that normalizations are attached to independent threshold systems, not to individual indicators.

Location normalization. The location of the ordinal measurement layer is not identified. One location normalization is therefore required for each independent threshold system.

A common approach is to fix one threshold, for example

$$\tau_{\ell 1} = 0,$$

which anchors the location of the latent response scale.

An alternative is to impose a centered threshold structure. When the response categories are themselves designed to be symmetric, as in many Likert scales (e.g., strongly disagree, disagree, neutral, agree, strongly agree), it may be reasonable to constrain the threshold system to be symmetric around zero. This specification simultaneously fixes the location of the threshold system and imposes an additional structural assumption reflecting the symmetry of the response scale. It also reduces the number of free parameters and often improves interpretability.

When symmetry is imposed, a convenient parameterization for $M_\ell = 5$ response categories is:

$$\begin{aligned}\tau_{\ell 1} &= -\delta_{\ell 0} - \delta_{\ell 1}, \\ \tau_{\ell 2} &= -\delta_{\ell 0}, \\ \tau_{\ell 3} &= \delta_{\ell 0}, \\ \tau_{\ell 4} &= \delta_{\ell 0} + \delta_{\ell 1},\end{aligned}$$

with $\delta_{\ell 0} > 0$ and $\delta_{\ell 1} > 0$. This guarantees strict ordering and centers the threshold system at zero, aligning the statistical specification with the semantics of balanced response scales.

Scale normalization. Ordered response models do not identify the scale of the latent response relative to threshold spacing and measurement noise. A scale normalization is therefore required for each independent threshold system.

For both ordered probit and ordered logit models, the disturbance term is assumed to follow a standardized distribution (respectively a standard normal or a standard logistic distribution). The scale of the latent response is then governed by the measurement scale parameter $\sigma_{v\ell}$. Because only relative scales are identified, one scale normalization must be imposed for each independent threshold system.

When several indicators share a common threshold system, this normalization is applied only once for the entire group. When different groups of indicators have distinct threshold systems, a separate scale normalization is required for each group.

2.4 Summary

The normalizations discussed above can be understood as fixing the *origins* and *units* of unobserved scales. Reference-indicator restrictions anchor each latent variable to an observed indicator, thereby fixing its location and scale. Separately, each ordered probit threshold system requires its own location and scale normalization, because only differences between the latent response and thresholds are identified.

Table 1 summarizes these requirements and makes explicit how often each restriction must be applied to avoid redundancy.

3 Hybrid choice models

This section builds on the notation and model components introduced in Section 1. We combine the structural equations for the latent variables, the measurement equations for the indicators (continuous or ordinal), and a discrete choice model, into two frameworks of increasing scope: the *MIMIC* model and the *hybrid choice model*.

Table 1: Summary of recommended normalization requirements in latent-variable and ordered probit models

Model component	Source of non-identification	Normalization required	Applied how often
Latent variable x_k^*	Location invariance	Fix one measurement intercept (reference indicator)	Once per latent variable
Latent variable x_k^*	Scale invariance	Fix one loading to ± 1 (reference indicator)	Once per latent variable
Continuous measurement equation	None beyond latent-variable invariance	No additional normalization	—
Ordinal measurement (ordered probit)	Location invariance	Fix one threshold to zero <i>or</i> impose symmetry	Once per independent threshold system
Ordinal measurement (ordered probit)	Scale invariance	Fix measurement error scale (e.g. $\sigma = 1$)	Once per independent threshold system

We use the same notations as in Section 1: $\mathbf{n}$ indexes individuals, $\mathbf{x}_{\mathbf{n}}$ denotes the observed explanatory variables, $\mathbf{x}_{\mathbf{n}}^*$ the vector of latent variables, $\mathbf{I}_{\mathbf{n}}$ the vector of observed indicators, and $\mathbf{i}_{\mathbf{n}} \in \mathcal{C}_{\mathbf{n}}$ the observed choice.

3.1 The MIMIC model

A *Multiple Indicators Multiple Causes* (MIMIC) model is obtained by combining:

- the structural part (“multiple causes”), which explains each latent variable as a function of observed covariates through the structural equations (2); and
- the measurement part (“multiple indicators”), which explains each indicator as a function of the latent variables through measurement equations (continuous indicators: (4); ordinal indicators: ordered probit, (12)).

More specifically, the latent variables are generated by the structural equations (1) (or their linear specification (2)), and they enter the measurement equations (3) or (11), depending on whether the indicators are continuous or ordinal.

The purpose of the MIMIC model is to infer latent variables from their observable manifestations (the indicators) while simultaneously explaining how these latent constructs vary with observed covariates.

As in the likelihood expressions derived in Section 1, we assume that the indicators are conditionally independent given the latent variables. For a given individual $\mathbf{n}$, the contribution of the indicators to the likelihood *conditional on the latent variables* $\mathbf{x}_{\mathbf{n}}^*$ is obtained by multiplying the indicator-specific contributions:

$$\Lambda_{\mathbf{n}}^{\mathbf{I}}(\mathbf{I}_{\mathbf{n}} \mid \mathbf{x}_{\mathbf{n}}^*, \mathbf{x}_{\mathbf{n}}; \lambda, \sigma_{\mathbf{v}}, \tau) = \prod_{\ell=1}^{L_{\mathbf{n}}} g_{\ell}(\mathbf{I}_{\mathbf{n}\ell} \mid \mathbf{x}_{\mathbf{n}}^*, \mathbf{x}_{\mathbf{n}}; \lambda_{\ell}, \sigma_{\mathbf{v}\ell}, \tau_{\ell}), \quad (16)$$

where

$$g_{\ell}(\cdot) = \begin{cases} \text{normal density implied by Eq. (4),} \\ \text{ordered-probit probability given by Eq. (12).} \end{cases}$$

For continuous indicators, g_{ℓ} depends on the measurement parameters λ_{ℓ} and the scale parameter $\sigma_{\mathbf{v}\ell}$. For ordinal indicators, it also depends on the threshold parameters τ_{ℓ} , or on the common threshold system τ when thresholds are shared across indicators.

Because $\mathbf{x}_{\mathbf{n}}^*$ is not observed, the individual likelihood contribution integrates the conditional indicator likelihood over the distribution of the latent variables implied by the structural equations:

$$\Lambda_{\mathbf{n}}^{\text{MIMIC}}(\alpha, \lambda, \sigma_{\mathbf{v}}, \tau) = \int_{\mathbf{x}_{\mathbf{n}}^*} \Lambda_{\mathbf{n}}^{\mathbf{I}}(\mathbf{I}_{\mathbf{n}} \mid \mathbf{x}_{\mathbf{n}}^*, \mathbf{x}_{\mathbf{n}}; \lambda, \sigma_{\mathbf{v}}, \tau) f(\mathbf{x}_{\mathbf{n}}^* \mid \mathbf{x}_{\mathbf{n}}; \alpha) d\mathbf{x}_{\mathbf{n}}^*, \quad (17)$$

where $f(\mathbf{x}_{\mathbf{n}}^* \mid \mathbf{x}_{\mathbf{n}}; \alpha)$ denotes the joint density of the latent variables implied by the structural equations (1)–(2).

This expression is the direct extension of the unconditional likelihoods derived in Eqs. (6) and the corresponding ordinal case.

3.2 Hybrid choice model (choice model with latent variables)

A *hybrid choice model* extends the MIMIC model by adding a discrete choice component in which latent variables enter the systematic utilities. The key modeling idea is that attitudes or perceptions (latent variables) may influence choices, while being measured only indirectly through the indicators.

In the notation introduced in Section 1, the latent variables $\mathbf{x}_n^*$ generated by the structural equations become additional explanatory variables in the utility specification.

Let $V(\mathbf{x}_{in}, \mathbf{x}_n^*; \boldsymbol{\beta})$ denote the systematic utility of alternative i for individual n . The probability of the observed choice i_n conditional on the latent variables is

$$P(i_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\beta}, \mu) = \frac{\exp(\mu V(\mathbf{x}_{in}, \mathbf{x}_n^*; \boldsymbol{\beta}))}{\sum_{j \in \mathcal{C}_n} \exp(\mu V(\mathbf{x}_{jn}, \mathbf{x}_n^*; \boldsymbol{\beta}))}, \quad (18)$$

where μ is the scale parameter of the logit model and $\boldsymbol{\beta}$ is the vector of utility parameters. Note that the logit model is adopted here for expositional convenience and because it is a widely used specification. However, the framework is fully general and can accommodate alternative discrete choice models, such as nested logit or cross-nested logit models, without any conceptual modification.

For a given individual n , the observed data consist of the choice i_n and the vector of indicators I_n . Conditional on $\mathbf{x}_n^*$, the hybrid choice model contribution to the likelihood is the product of:

- the conditional choice probability from Eq. (18),
- the conditional indicator likelihood from Eq. (16).

That is,

$$\Lambda_n(i_n, I_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\beta}, \mu, \boldsymbol{\lambda}, \sigma_v, \boldsymbol{\tau}) = P(i_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\beta}, \mu) \Lambda_n^I(I_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\lambda}, \sigma_v, \boldsymbol{\tau}). \quad (19)$$

This factorization relies on the assumption that, conditional on the latent variables, the choice and the indicators are independent.

Because $\mathbf{x}_n^*$ is latent, the individual likelihood contribution integrates (19) over $f(\mathbf{x}_n^* | \mathbf{x}_n; \boldsymbol{\alpha})$ implied by the structural equations (2):

$$\Lambda_n^{\text{HCM}}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \mu, \boldsymbol{\lambda}, \sigma_v, \boldsymbol{\tau}) = \int_{\mathbf{x}_n^*} P(i_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\beta}, \mu) \Lambda_n^I(I_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\lambda}, \sigma_v, \boldsymbol{\tau}) f(\mathbf{x}_n^* | \mathbf{x}_n; \boldsymbol{\alpha}) d\mathbf{x}_n^*. \quad (20)$$

Let $\boldsymbol{\theta}$ denote the full parameter vector, including the utility parameters $\boldsymbol{\beta}$, the scale parameter μ , the structural parameters $\boldsymbol{\alpha}$, the measurement parameters $\boldsymbol{\lambda}$, the threshold parameters $\boldsymbol{\tau}$ for ordinal indicators, and any scale parameters. The parameter vector $\boldsymbol{\theta}$ is assumed to satisfy the normalization conditions discussed in Section 2.

The sample log-likelihood is

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}) &= \sum_n \ln \Lambda_n^{\text{HCM}}(\boldsymbol{\alpha}, \boldsymbol{\beta}, \mu, \boldsymbol{\lambda}, \sigma_v, \boldsymbol{\tau}) \\ &= \sum_n \ln \left[\int_{\mathbf{x}_n^*} P(i_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\beta}, \mu) \Lambda_n^I(I_n | \mathbf{x}_n^*, \mathbf{x}_n; \boldsymbol{\lambda}, \sigma_v, \boldsymbol{\tau}) f(\mathbf{x}_n^* | \mathbf{x}_n; \boldsymbol{\alpha}) d\mathbf{x}_n^* \right]. \end{aligned} \quad (21)$$

The integral in (21) typically has no closed form and is evaluated by Monte-Carlo integration.

4 A case study

This case study illustrates, step by step, how the modeling ingredients introduced in Sections 1, 2, and 3 can be implemented with Biogeme. Rather than presenting directly the most general hybrid specification, we adopt a progressive approach. At each step, only one modeling dimension is modified, so that the role of each component can be understood clearly: the choice model, the latent variables, the measurement equations, the estimation strategy, and the implementation style.

The empirical application is based on revealed-preference data collected in Switzerland in the context of the OPTIMA project (Bierlaire et al., 2011). The survey was conducted between 2009 and 2010 on behalf of CarPostal, the public transport operator of the Swiss Postal Service. Its objective was to better understand travel behavior in low-density areas, which represent the typical service environment of CarPostal. In addition to observed mode choices and explanatory variables, the survey contains several psychometric indicators that can be used to infer latent constructs. The data file and its description are available on the Biogeme webpage. A description of the variables is also provided in Appendix A.

The sequence of examples is organized as a set of increasingly rich specifications. We start with a standard discrete choice model without latent variables, which serves as a reference specification. We then introduce latent variables in isolation through a MIMIC model, using the continuous case for the measurement equations in order to focus on the structural equation, the measurement equation, and the associated normalization issues. Once this foundation is established, the latent variable is incorporated into a mode choice model, first using a sequential estimation strategy and then using simultaneous estimation. This allows us to contrast a simplified estimation approach with the full hybrid formulation described in Section 3.

In a second stage, we replace the continuous measurement equations by ordinal measurement equations. This is an important step because the psychometric indicators available in the data are naturally interpreted as ordinal responses rather than as continuous measurements. Two variants are considered: ordered logit and ordered probit. The goal is not to change the behavioral content of the model, but to show how the same latent-variable and choice structure can be combined with different ordinal measurement schemes and with the corresponding normalization of threshold systems.

We now present the examples in the order described above.

4.1 Example 1: baseline mode choice model

We start with the simplest specification considered in this tutorial, a standard logit model for mode choice, without explicit latent variables. Its role is to provide a reference specification against which the richer models introduced later can be compared.

For each individual $\mathbf{n}$, let $\mathcal{C}_{\mathbf{n}}$ denote the set of available transport modes, and let $i_{\mathbf{n}} \in \mathcal{C}_{\mathbf{n}}$ be the observed mode choice. In the application, three alternatives are considered:

- public transport,
- car,
- slow modes.

Their systematic utilities depend only on observed explanatory variables. Using the notation introduced in Section 3.2, they are specified as follows.

For public transport,

$$V_{PT,\mathbf{n}} = \beta_{asc,PT} + \beta_{time,PT} \text{TimePT}_{\mathbf{n}} + \beta_{cost} \text{MarginalCostPT}_{\mathbf{n}}. \quad (22)$$

For car,

$$V_{Car,\mathbf{n}} = \beta_{asc,Car} + \beta_{time,Car} \text{TimeCar}_{\mathbf{n}} + \beta_{cost} \text{CostCar}_{\mathbf{n}}. \quad (23)$$

For slow modes,

$$V_{Slow,\mathbf{n}} = \beta_{dist,\mathbf{n}} \text{Distance}_{\mathbf{n}}, \quad (24)$$

where the distance coefficient is specified as a function of the trip purpose:

$$\beta_{dist,\mathbf{n}} = \beta_{dist,work} \mathbf{1}\{\text{PurPWH}_{\mathbf{n}} = 1\} + \beta_{dist,other} \mathbf{1}\{\text{PurPWH}_{\mathbf{n}} \neq 1\}. \quad (25)$$

In these expressions, TimePT_n , TimeCar_n , MarginalCostPT_n , CostCar_n , and Distance_n denote the observed travel attributes for individual n , while $\mathbf{1}\{\cdot\}$ is the indicator function selecting work trips from all other trip purposes.

As discussed in Section 2, discrete choice models require normalization to ensure identification. In the present specification, the location of the utility scale is fixed by setting the alternative-specific constant associated with slow modes to zero. This normalization is implicit in the specification (24), where no constant term is included for slow modes.

The scale of utility is identified by normalizing the cost coefficient to

$$\beta_{\text{cost}} = -1.$$

Indeed, only relative utility differences are identified in discrete choice models, and multiplying all utility coefficients by a common positive constant leaves the choice probabilities unchanged up to a corresponding rescaling of the logit scale parameter. Fixing the cost coefficient therefore defines the unit in which utility is measured.

With this normalization in place, the scale parameter appearing in the logit model can be estimated. The systematic utilities are multiplied by a positive scale parameter μ .

At this stage, the systematic utilities depend only on observed variables. No additional latent construct $\mathbf{x}_n^*$ is introduced in the specification. The objective of this first example is therefore to isolate the choice component before augmenting it later with structural equations for latent variables and with psychometric indicators.

Under the logit specification introduced in Eq. (18), the probability of the observed choice is

$$P(i_n | \mathbf{x}_n; \beta) = \frac{\exp(\mu V(\mathbf{x}_{i_n, n}; \beta))}{\sum_{j \in c_n} \exp(\mu V(\mathbf{x}_{j, n}; \beta))}, \quad (26)$$

where μ is the scale parameter of the logit model and β is the vector collecting all utility parameters.

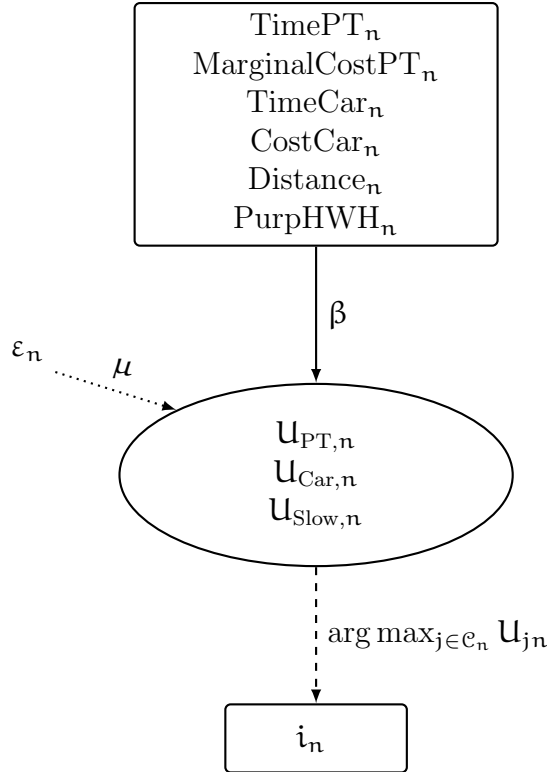


Figure 1: Graphical representation of the baseline mode choice model.

Figure 1 provides a graphical representation of the baseline mode choice model.

The observed explanatory variables $\mathbf{x}_n$, listed in the upper node, enter the specification of the random utility functions $U_{PT,n}$, $U_{Car,n}$, and $U_{Slow,n}$ through their deterministic component. This corresponds to the structural specification of the model, where the systematic part of utility is constructed as a function of observed attributes and parameters β .

The random term ε_n captures unobserved factors affecting utility. It enters the utility functions with a scale parameter μ , reflecting the normalization of the variance of the error terms.

The lower node represents the observed choice i_n , which is linked to the latent utilities through the utility-maximization rule. The dashed arrow captures this measurement relation: the observed choice reveals which alternative attains the maximum utility. Under the assumption that the error terms are i.i.d. extreme value, this behavioral rule leads to the logit probability model defined in Eq. (18).

Because no additional latent variable $\mathbf{x}_n^*$ is introduced in this first example, there is no structural equation of the form (2) describing latent constructs beyond the utility functions themselves, and no psychometric indicators contributing through the conditional indicator likelihood (16). As a consequence, the individual contribution to the likelihood reduces to the choice component alone:

$$\Lambda_n(\beta) = P(i_n | \mathbf{x}_n; \beta). \quad (27)$$

The sample log-likelihood is therefore

$$\mathcal{L}(\beta) = \sum_n \ln P(i_n | \mathbf{x}_n; \beta). \quad (28)$$

This specification is implemented in the script `plot_h01_mode_logit.py`. The corresponding utility functions are generated in the module

`choice_latent_variables.py`.

Although that module is written so that latent-variable terms can be added later, these terms are not activated in the present example.

This first example is deliberately simple. It establishes the choice component of the case study before introducing, in the following examples, explicit latent variables, structural equations, indicator equations, normalization issues, and the Monte-Carlo integration required to evaluate the hybrid choice model likelihood (21).

The estimation results reported in Table 2 are consistent with expectations. The estimated coefficients associated with travel time and distance are all negative and statistically significant, indicating that longer travel times and greater distances reduce utility. Because the cost coefficient is fixed to -1 , the time coefficients can be interpreted as values of time expressed in monetary units. Their magnitudes suggest a larger value of time for car users than for public transport users. Distance has a significant effect on the utility of slow modes, with a stronger impact for non-work trips than for work trips. Finally, the estimated scale parameter is positive and statistically significant, confirming the validity of the chosen normalization.

Number of estimated parameters	7
Sample size	889
Init log likelihood	-5937.274
Final log likelihood	-508.6166
Akaike Information Criterion	1031.233
Bayesian Information Criterion	1064.764
Optimization time	0:00:01.416986

Description	Coeff. estimate	Robust Asympt.		t-stat	p-value
		std.	error		
choice_scale_parameter	0.0782	0.0187		4.18	2.97e-05
choice_asc_car	-3.07	5.03		-0.609	0.542
choice_asc_pt	-8.71	5.90		-1.48	0.140
choice_beta_time_pt	-12.2	4.26		-2.86	0.00424
choice_beta_time_car	-26.0	8.84		-2.94	0.00328
choice_beta_dist_work	-2.61	0.840		-3.11	0.00186
choice_beta_dist_other_purposes	-4.12	1.22		-3.38	0.000732

Table 2: Estimation results for the baseline mode choice model

4.2 Example 2: MIMIC model in the continuous case

We now introduce the first latent-variable specification of the case study. In contrast with Example 4.1, this model contains no choice component. It is therefore a pure *Multiple Indicators Multiple Causes* (MIMIC) model, as described in Section 3.1. Its purpose is to introduce the latent variable, its structural equation, the associated measurement equations, and the normalization strategy, without the additional complexity of a choice model.

In this example, a single latent variable is considered, $\mathbf{x}_{n,\text{car}}^*$, interpreted as a *car-centric attitude*. The structural equation follows the linear specification of Eq. (2):

$$\begin{aligned} \mathbf{x}_{n,\text{car}}^* = & \psi_0 + \psi_1 \text{HighEducation}_n + \psi_2 \text{TopManager}_n \\ & + \psi_3 \text{CarOrientedParents}_n + \psi_4 \text{LowEducation}_n \\ & + \psi_5 \text{UsedToGoToSchoolByCar}_n + \sigma_{\omega,\text{car}} \omega_n. \end{aligned} \quad (29)$$

where $\omega_n \sim \mathcal{N}(0, 1)$. The explanatory variables included in the structural equation are exactly those declared in the semantic specification used by the script: `top_manager`, `age_30_less`, `car_oriented_parents`, `high_education`, `low_education`, and `used_to_go_to_school_by_car`. As in the implementation, the error scale $\sigma_{\omega,\text{car}}$ is estimated and constrained to be positive.

The latent variable $\mathbf{x}_{n,\text{car}}^*$ is measured using the following Likert indicators:

Envir01 Fuel price should be increased to reduce congestion and air pollution.

Envir02 More public transportation is needed, even if taxes are set to pay the additional costs.

Envir06 Actions and decision making are needed to limit greenhouse gas emissions.

Mobil03 I use the time of my trip in a productive way.

Mobil05 I reconsider frequently my mode choice.

Mobil08 I do not feel comfortable when I travel close to people I do not know.

Mobil09 Taking the bus helps making the city more comfortable and welcoming.

Mobil10 It is difficult to take the public transport when I travel with my children.

LifSty07 The pleasure of having something beautiful consists in showing it.

All these indicators are treated as ordinal variables defined on a common symmetric Likert scale with five categories. They provide indirect observations of the latent construct $\mathbf{x}_{n,\text{car}}^*$, which is not directly observed and must be inferred through the measurement equations introduced in Section 1.3.

Although these indicators are observed on ordinal response scales, they are treated here according to the *continuous case* introduced in Section 1.2. This is a deliberate simplification for pedagogical purposes: it allows us to focus first on the latent variable itself and on its identification, before introducing ordinal measurement equations in later examples.

For each indicator ℓ in the above set, the measurement equation is written as

$$I_{n\ell} = \lambda_{\ell 0} + \lambda_{\ell 1} \mathbf{x}_{n,\text{car}}^* + \sigma_{v\ell} \mathbf{v}_{n\ell}, \quad (30)$$

where $\mathbf{v}_{n\ell} \sim \mathcal{N}(0, 1)$. Therefore, conditional on the latent variable, the contribution of the indicators to the likelihood is

$$\Lambda_n^I(I_n | \mathbf{x}_{n,\text{car}}^*, \mathbf{x}_n) = \prod_{\ell=1}^{L_n} \frac{1}{\sigma_{v\ell}} \phi\left(\frac{I_{n\ell} - \lambda_{\ell 0} - \lambda_{\ell 1} \mathbf{x}_{n,\text{car}}^*}{\sigma_{v\ell}}\right), \quad (31)$$

where $\phi(\cdot)$ denotes the standard normal density.

Because the latent variable is not observed, the likelihood contribution of individual $\mathbf{n}$ is obtained by integrating out $\mathbf{x}_{n,\text{car}}^*$:

$$\Lambda_n^{\text{MIMIC}} = \int_{\mathbf{x}_{n,\text{car}}^*} \Lambda_n^I(I_n | \mathbf{x}_{n,\text{car}}^*, \mathbf{x}_n) f(\mathbf{x}_{n,\text{car}}^* | \mathbf{x}_n) d\mathbf{x}_{n,\text{car}}^*, \quad (32)$$

where $f(\mathbf{x}_{n,\text{car}}^* | \mathbf{x}_n; \boldsymbol{\alpha})$ is the conditional density implied by the structural equation (29). As discussed in Section 1.2, this integral has no closed form and is evaluated by Monte-Carlo integration.

Normalization. As explained in Section 2, identification of the latent variable requires fixing its location and its scale. In the implementation, this is achieved by adopting a reference-indicator strategy based on indicator `Envir01`. Its intercept is fixed to zero:

$$\lambda_{\text{Envir01},0} = 0, \quad (33)$$

which anchors the location of the latent-variable scale. Its loading on the latent variable is fixed to -1 :

$$\lambda_{\text{Envir01},1} = -1, \quad (34)$$

which anchors the scale and the orientation of the latent variable. With this sign convention, larger values of the car-centric attitude correspond to lower expected values of `Envir01`. Once these two constraints are imposed, the remaining intercepts and loadings are interpreted relative to the reference indicator.

This specification is implemented in the script `plot_h02_lv_mimic_gauss.py`. The semantic specification of the latent variable itself is defined separately in `one_latent_variable_spec.py`; the example script then adds the continuous measurement equations, the normalization constraints, and the maximum-likelihood estimation setup. The resulting model is the first example in the tutorial where the latent-variable machinery is used explicitly.

Description	BHHH				
	Coeff.	Asympt.			
	estimate	std.	error	t-stat	p-value

measurement_Envir01_sigma_log	0.00969	0.0450	0.215	0.829
measurement_intercept_Envir02	1.76	0.160	11.0	0.00
measurement_coefficient_car_ centric_attitude_Envir02	-0.597	0.0574	-10.4	0.00
measurement_Envir02_sigma_log	0.0247	0.0271	0.910	0.363
measurement_intercept_Envir06	3.30	0.101	32.8	0.00
measurement_coefficient_car_ centric_attitude_Envir06	-0.376	0.0385	-9.77	0.00
measurement_Envir06_sigma_log	-0.303	0.0183	-16.5	0.00
measurement_intercept_Mobil03	2.86	0.156	18.3	0.00
measurement_coefficient_car_ centric_attitude_Mobil03	-0.150	0.0550	-2.73	0.00629
measurement_Mobil03_sigma_log	0.289	0.0286	10.1	0.00
measurement_intercept_Mobil05	2.42	0.172	14.0	0.00
measurement_coefficient_car_ centric_attitude_Mobil05	-0.339	0.0607	-5.59	2.25e-08
measurement_Mobil05_sigma_log	0.320	0.0288	11.1	0.00
measurement_intercept_Mobil08	3.22	0.150	21.4	0.00
measurement_coefficient_car_ centric_attitude_Mobil08	0.336	0.0550	6.11	9.73e-10
measurement_Mobil08_sigma_log	0.141	0.0255	5.52	3.36e-08
measurement_intercept_Mobil09	2.95	0.133	22.1	0.00
measurement_coefficient_car_ centric_attitude_Mobil09	-0.351	0.0498	-7.05	1.80e-12
measurement_Mobil09_sigma_log	0.0575	0.0196	2.94	0.00329
measurement_intercept_Mobil10	5.30	0.266	19.9	0.00
measurement_coefficient_car_ centric_attitude_Mobil10	0.533	0.0896	5.95	2.73e-09
measurement_Mobil10_sigma_log	0.594	0.0428	13.9	0.00
measurement_intercept_LifSty07	2.49	0.141	17.7	0.00
measurement_coefficient_car_ centric_attitude_LifSty07	0.0866	0.0488	1.77	0.0761
measurement_LifSty07_sigma_log	0.217	0.0247	8.76	0.00

Table 5: Estimation results for the MIMIC model (continuous measurement): measurement equations

The estimation results for the structural equation are reported in Table 4. The estimated intercept is negative and highly significant, indicating a low average level of the car-centric attitude in the population under the chosen normalization. Among the explanatory variables, having car-oriented parents is associated with a significantly higher car-centric attitude, whereas a high level of education is associated with a significantly lower one. The remaining variables, namely `top_manager`, `low_education`, and `used_to_go_to_school_by_car`, are not statistically significant in this specification. The estimated scale parameter of the structural disturbance is close to one, as $\exp(-0.00271) \approx 1$, indicating that a substantial share of the variability of the latent attitude remains unexplained by the observed socio-demographic characteristics.

The estimation results for the measurement equations are reported in Table 5. With the exception of `LifSty07`, all indicator loadings are statistically significant and therefore provide information about the latent construct. The signs of the coefficients are consistent with the normalization adopted through the reference indicator `Envir01`: larger values of the car-centric attitude are associated with lower expected values of the environmental and public-transport-related indicators (`Envir02`, `Envir06`, `Mobil05`, and `Mobil09`), while they are associated with higher expected values of indicators reflecting greater discomfort with public transport or a stronger attachment to private mobility (`Mobil08` and `Mobil10`). The loading of `Mobil03`

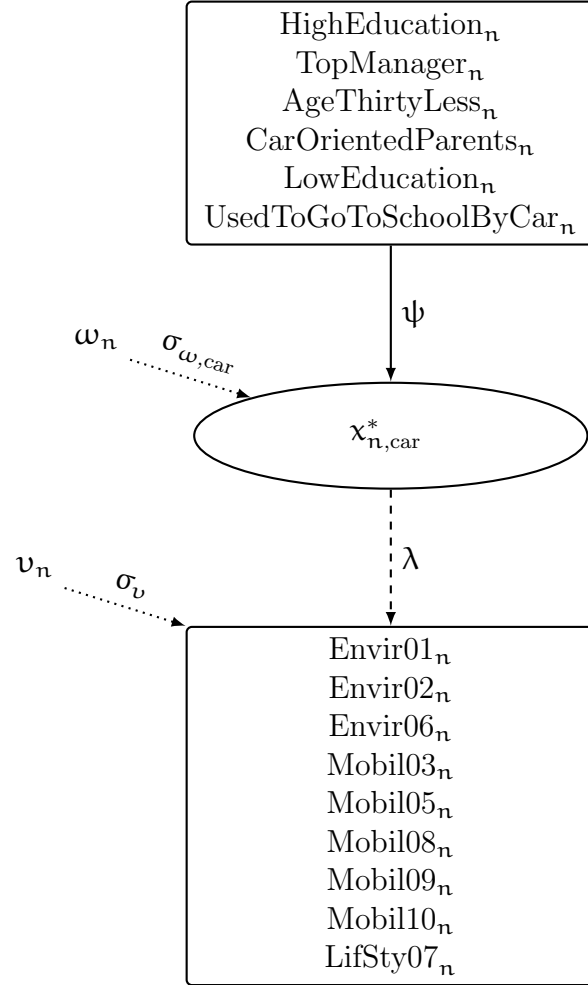


Figure 2: Graphical representation of the MIMIC model for the car-centric latent variable.

Number of estimated parameters	32
Sample size	889
Init log likelihood	-26577.53
Final log likelihood	-13029.98
Akaike Information Criterion	26123.95
Bayesian Information Criterion	26277.24
Number of draws	50000
Optimization time	0:25:49.527588

Table 3: Estimation results for the MIMIC model (continuous measurement): summary statistics

Description	Coeff. estimate	Asympt. std. error	t-stat	p-value
struct_car_centric_attitude_ intercept	-2.64	0.115	-23.0	0.00
struct_car_centric_attitude_ top_manager	0.0914	0.143	0.638	0.523
struct_car_centric_attitude_ car_oriented_parents	0.237	0.102	2.32	0.0203
struct_car_centric_attitude_ high_education	-0.541	0.130	-4.15	3.27e-05
struct_car_centric_attitude_ low_education	0.123	0.125	0.983	0.326
struct_car_centric_attitude_ used_to_go_to_school_by_car	0.0998	0.560	0.178	0.859
struct_car_centric_attitude_ sigma_log	-0.00271	0.0740	-0.0366	0.971

Table 4: Estimation results for the MIMIC model (continuous measurement): structural equation

is statistically significant but comparatively small, suggesting a weaker relationship with the latent construct. Overall, the estimated loadings confirm that the indicators capture a common underlying attitudinal dimension that can reasonably be interpreted as a car-centric attitude.

4.3 Example 3: hybrid mode choice model with sequential estimation

We now extend the baseline mode choice model by introducing the latent variable estimated in Section 4.2 into the utility specification. This is our first hybrid choice model, estimated using a *sequential* approach.

Let $\mathbf{x}_{n,\text{car}}^*$ denote the car-centric latent variable introduced in Section 4.2. As in the selected implementation, its structural equation is not re-estimated jointly with the choice model. Instead, the structural parameters estimated in the MIMIC model are treated as fixed and the latent variable is specified as

$$\begin{aligned}
\mathbf{x}_{n,\text{car}}^* = & \hat{\psi}_0 + \hat{\psi}_1 \text{HighEducation}_n + \hat{\psi}_2 \text{TopManager}_n + \hat{\psi}_3 \text{AgeThirtyLess}_n \\
& + \hat{\psi}_4 \text{CarOrientedParents}_n + \hat{\psi}_5 \text{LowEducation}_n + \hat{\psi}_6 \text{UsedToGoToSchoolByCar}_n \\
& + \hat{\sigma}_{\omega,\text{car}} \omega_n,
\end{aligned} \tag{35}$$

where $\omega_n \sim \text{N}(0, 1)$. Therefore, although the structural parameters are fixed, the latent variable remains random through the structural disturbance term.

The systematic utilities are then augmented with this latent variable. The specifications for public transport and slow modes remain unchanged with respect to Example 4.1. For the car

alternative, the utility function explicitly incorporates the latent variable as follows:

$$V_{\text{Car},n} = \beta_{\text{asc},\text{Car}} + \beta_{\text{time},\text{Car}} \text{TimeCar}_n + \beta_{\text{cost}} \text{CostCar}_n + \beta_{\text{car,LV}} x_{n,\text{car}}^*, \quad (36)$$

where $\beta_{\text{car,LV}}$ captures the impact of the car-centric attitude on the utility of the car alternative.

The conditional choice probability keeps the standard logit form:

$$P(i_n | x_n, x_{n,\text{car}}^*; \beta) = \frac{\exp(V(x_{i_n,n}, x_{n,\text{car}}^*; \beta))}{\sum_{j \in \mathcal{C}_n} \exp(V(x_{j,n}, x_{n,\text{car}}^*; \beta))}. \quad (37)$$

Because the latent variable is random, the individual likelihood contribution involves integration over its distribution:

$$\Lambda_n(\beta) = \int_{x_{n,\text{car}}^*} P(i_n | x_n, x_{n,\text{car}}^*; \beta) f(x_{n,\text{car}}^* | x_n; \alpha) dx_{n,\text{car}}^*, \quad (38)$$

and the log-likelihood is

$$\mathcal{L}(\beta) = \sum_n \ln \left[\int_{x_{n,\text{car}}^*} P(i_n | x_n, x_{n,\text{car}}^*; \beta) f(x_{n,\text{car}}^* | x_n; \alpha) dx_{n,\text{car}}^* \right]. \quad (39)$$

In this specification, the indicators used in the previous example do not appear explicitly in the likelihood. They intervene only indirectly through the previously estimated structural parameters $\hat{\psi}$. This is the defining feature of the sequential approach: the latent variable is first estimated from the measurement model, and then introduced into the choice model without joint estimation.

This model is implemented in the script `plot_h03_model_lv_gauss_seq.py`. The script reads the results of the MIMIC model, reconstructs the latent variable, inserts it into the utility functions, and estimates the choice model by maximum likelihood.

While this approach is simple and useful for illustration, it does not correspond to the full hybrid likelihood of Eq. (21), as the measurement equations are not estimated jointly with the choice model. The next example will address this limitation by moving to simultaneous estimation.

Figure 3 provides a graphical representation of the sequential hybrid choice model with one latent variable. The figure combines the choice model (left) and the latent variable model (right), together with the link between them.

On the left-hand side, the observed explanatory variables x_n , listed in the upper node, enter the specification of the random utility functions $U_{\text{PT},n}$, $U_{\text{Car},n}$, and $U_{\text{Slow},n}$. This corresponds to the structural part of the choice model, with parameters β . The random utilities are also affected by an unobserved disturbance ε_n , scaled by the parameter μ . The observed choice i_n results from the selection of the alternative with the highest utility, represented by the mapping $i_n = \arg \max_{j \in \mathcal{C}_n} U_{j,n}$.

On the right-hand side, the latent variable $x_{n,\text{car}}^*$ is specified through a structural equation as a function of observed characteristics, with parameters $\hat{\psi}$. It is also affected by a random term ω_n , scaled by $\hat{\sigma}_{\omega,\text{car}}$. In the sequential estimation approach, these parameters are estimated in a first step using the measurement model and are treated as fixed in the subsequent choice model estimation.

The horizontal link between the latent variable and the utility functions captures the role of attitudes in the choice process. The latent variable enters the utility specification through the parameter β_{LV} , thereby influencing the relative attractiveness of the alternatives. This representation highlights the two-stage nature of the sequential approach: the latent variable is first constructed from its structural equation, and then incorporated as an explanatory factor in the choice model.

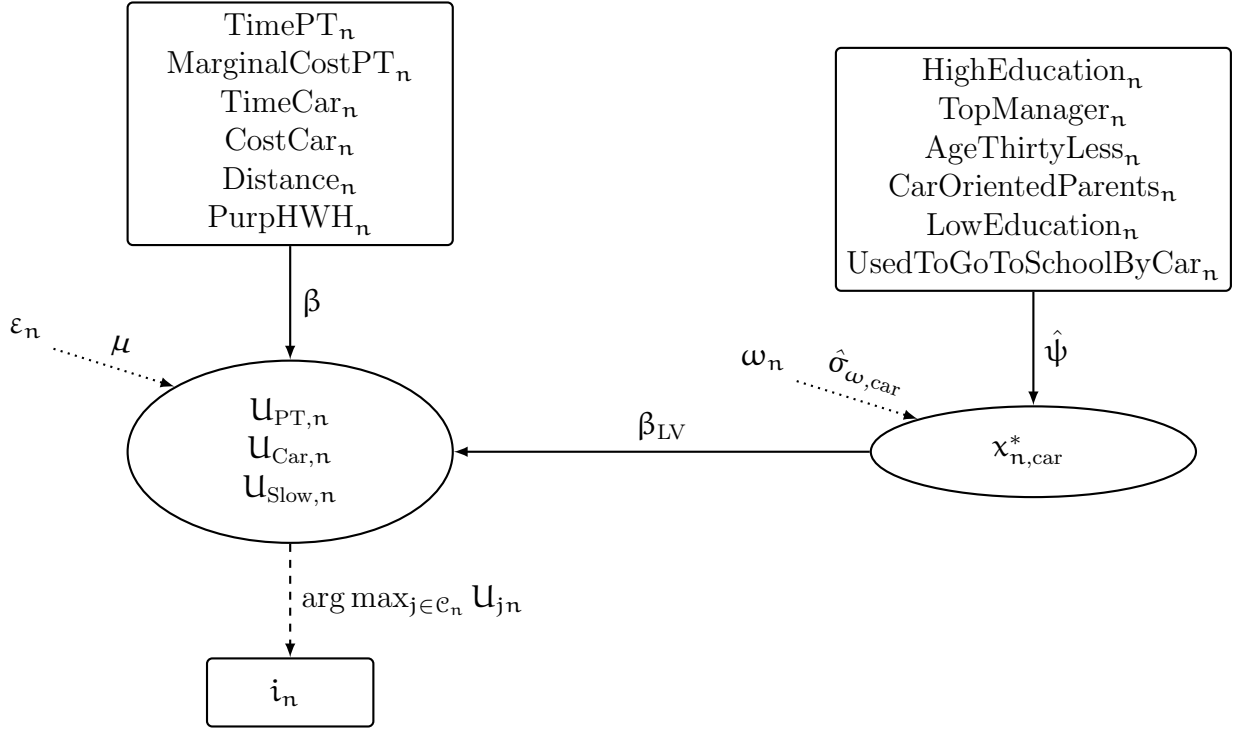


Figure 3: Graphical representation of the sequential hybrid choice model with one latent variable.

Number of estimated parameters	8
Sample size	889
Init log likelihood	-5607.637
Final log likelihood	-507.5701
Akaike Information Criterion	1031.14
Bayesian Information Criterion	1069.461
Number of draws	50000
Optimization time	0:00:53.058620

Table 6: Estimation results for the hybrid choice model with sequential estimation: summary statistics

The estimation results are reported in Table 7. The coefficients associated with travel time and distance remain remarkably stable compared to those obtained for the baseline mode choice model in Table 2. They retain the same signs, magnitudes, and levels of statistical significance, indicating that the introduction of the latent variable does not substantially modify the effects of the observed travel attributes. As in the baseline model, the implied value of time is larger for car than for public transport, and distance has a stronger impact on the utility of slow modes for non-work trips than for work trips. The coefficient associated with the car-centric latent variable is positive, implying that individuals with a stronger car-oriented attitude tend to derive greater utility from the car alternative. However, this effect is only marginally significant ($p = 0.101$), providing limited statistical evidence that the latent attitude contributes to explaining mode choice beyond the observed socio-demographic and travel attributes. The estimated scale parameter remains virtually unchanged from the baseline specification.

Description	Coeff. estimate	Robust Asympt.		t-stat	p-value
		std.	error		
choice_scale_parameter	0.0782	0.0190		4.12	3.80e-05
choice_asc_pt	-8.79	5.95		-1.48	0.140
choice_beta_time_pt	-12.3	4.34		-2.84	0.00457
choice_asc_car	-2.65	5.04		-0.527	0.598
choice_beta_time_car	-26.2	9.01		-2.91	0.00361
choice_beta_car_centric_	3.80	2.32		1.64	0.101
attitude_car					
choice_beta_dist_work	-2.62	0.845		-3.10	0.00195
choice_beta_dist_other_	-4.14	1.24		-3.35	0.000813
purposes					

Table 7: Estimation results for the hybrid choice model with sequential estimation

4.4 Example 4: hybrid mode choice model with simultaneous estimation

We now move from the sequential approach of Example 4.3 to the full hybrid specification described in Section 3.2. The behavioral content of the model is unchanged: we consider the same mode choice model, the same car-centric latent variable, the same structural equation, and the same continuous measurement equations as in Section 4.2. What changes is the estimation strategy. Instead of estimating the latent-variable model first and then introducing the latent variable into the choice model, all components are now estimated jointly.

The latent variable $x_{n,car}^*$ is specified exactly as in Section 4.2, with the same structural equation and the same normalization. Likewise, the measurement equations of the indicators are the continuous measurement equations introduced in Section 1.2 and used in Example 4.2. The utility specification is the same as in Example 4.3, in particular for the car alternative:

$$V_{Car,n} = \beta_{asc,Car} + \beta_{time,Car} \text{TimeCar}_n + \beta_{cost} \text{CostCar}_n + \beta_{car,LV} x_{n,car}^*. \quad (40)$$

The key difference is that the latent-variable parameters and the choice-model parameters are now estimated in one single optimization problem.

For each individual n , the full observation is the pair (i_n, I_n) , where i_n is the observed mode choice and I_n is the vector of psychometric indicators. Conditional on the latent variable, the individual contribution to the likelihood is therefore the product of the conditional choice probability and the conditional indicator likelihood:

$$\Lambda_n(i_n, I_n | x_n^*, x_n; \beta, \lambda, \tau) = P(i_n | x_n^*, x_n; \beta) \Lambda_n^I(I_n | x_n^*, x_n; \lambda, \tau), \quad (41)$$

where the first term is the logit choice probability and the second term is the continuous measurement contribution already introduced in Section 4.2. Because the latent variable is not observed, the individual likelihood contribution becomes

$$\Lambda_n = \int_{x_{n,car}^*} P(i_n | x_n, x_{n,car}^*; \beta) \Lambda_n^I(I_n | x_{n,car}^*, x_n) f(x_{n,car}^* | x_n; \alpha) dx_{n,car}^*, \quad (42)$$

which is exactly the hybrid choice model likelihood specialized to one latent variable and continuous measurement equations. As before, the integral is evaluated by Monte-Carlo integration.

Figure 4 provides a graphical representation of this model. The left-hand side corresponds to the choice component: observed explanatory variables enter the latent utilities, which in turn determine the observed choice. The right-hand side corresponds to the latent-variable model: observed socio-demographic variables determine the latent variable $x_{n,car}^*$, which is measured

Number of estimated parameters	40
Sample size	889
Init log likelihood	-32514.8
Final log likelihood	-13512.63
Akaike Information Criterion	27105.25
Bayesian Information Criterion	27296.86
Number of draws	50000
Optimization time	1:29:34.924149

Table 8: Estimation results for the simultaneous hybrid choice model (one latent variable, continuous measurement): summary statistics

through the vector of observed indicators. The horizontal arrow linking $x_{n,car}^*$ to the utility system emphasizes the defining feature of the hybrid model: the same latent construct is both measured through the psychometric indicators and used as an explanatory factor in the choice model. Unlike the sequential approach, the figure represents here a truly joint model, in the sense that the two blocks contribute simultaneously to the same likelihood function.

This specification is implemented in the script `plot_h04_model_lv_gauss_simult.py`. The script resolves the semantic specification of the latent variable, builds the continuous measurement model, constructs the mode-choice utilities including the latent-variable term, multiplies the conditional measurement and choice likelihood contributions, and integrates the result over the latent variable. This is the first example in the tutorial where the full hybrid likelihood is estimated directly.

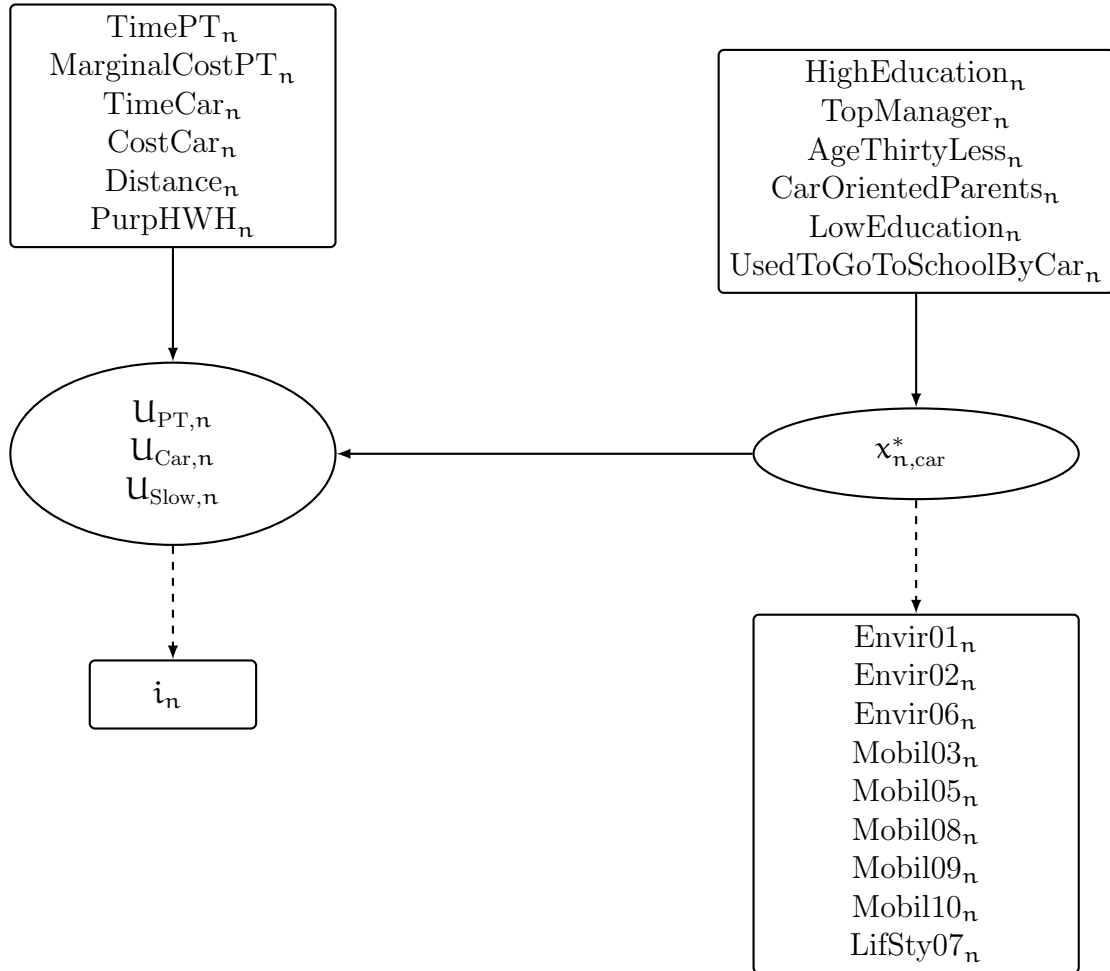


Figure 4: Graphical representation of the hybrid choice model with one latent variable.

Description	Coeff. estimate	BHHH Asympt.		t-stat	p-value
		std.	error		
struct_car_centric_attitude_ intercept	-2.66	0.116		-22.9	0.00
struct_car_centric_attitude_ top_manager	0.123	0.145		0.852	0.394
struct_car_centric_attitude_ car_oriented_parents	0.268	0.104		2.57	0.0102
struct_car_centric_attitude_ high_education	-0.530	0.132		-4.01	6.11e-05
struct_car_centric_attitude_ low_education	0.122	0.127		0.961	0.336
struct_car_centric_attitude_ used_to_go_to_school_by_car	0.119	0.550		0.217	0.828
struct_car_centric_attitude_ sigma_log	0.0217	0.0725		0.300	0.764

Table 9: Estimation results for the simultaneous hybrid choice model (one latent variable, continuous measurement): structural equation

Description	Coeff. estimate	BHHH Asympt.		t-stat	p-value
		std.	error		
measurement_Envir01_sigma_log	-0.0157	0.0469		-0.335	0.737
measurement_intercept_Envir02	1.81	0.154		11.7	0.00
measurement_coefficient_car_ centric_attitude_Envir02	-0.581	0.0554		-10.5	0.00
measurement_Envir02_sigma_log	0.0262	0.0276		0.950	0.342
measurement_intercept_Envir06	3.35	0.0956		35.1	0.00
measurement_coefficient_car_ centric_attitude_Envir06	-0.358	0.0369		-9.70	0.00
measurement_Envir06_sigma_log	-0.295	0.0174		-17.0	0.00
measurement_intercept_Mobil03	2.83	0.153		18.5	0.00
measurement_coefficient_car_ centric_attitude_Mobil03	-0.163	0.0538		-3.04	0.00240
measurement_Mobil03_sigma_log	0.287	0.0289		9.94	0.00
measurement_intercept_Mobil05	2.40	0.167		14.4	0.00
measurement_coefficient_car_ centric_attitude_Mobil05	-0.345	0.0585		-5.89	3.85e-09
measurement_Mobil05_sigma_log	0.317	0.0290		11.0	0.00
measurement_intercept_Mobil08	3.16	0.144		21.9	0.00
measurement_coefficient_car_ centric_attitude_Mobil08	0.314	0.0531		5.92	3.19e-09
measurement_Mobil08_sigma_log	0.145	0.0257		5.65	1.58e-08
measurement_intercept_Mobil09	2.99	0.128		23.3	0.00
measurement_coefficient_car_ centric_attitude_Mobil09	-0.336	0.0480		-7.00	2.60e-12
measurement_Mobil09_sigma_log	0.0601	0.0195		3.08	0.00204
measurement_intercept_Mobil10	5.28	0.260		20.3	0.00
measurement_coefficient_car_ centric_attitude_Mobil10	0.524	0.0874		6.00	1.99e-09
measurement_Mobil10_sigma_log	0.593	0.0429		13.8	0.00
measurement_intercept_LifSty07	2.42	0.139		17.5	0.00

Description	Coeff. estimate	BHHH		p-value
		Asympt. std. error	t-stat	
choice_scale_parameter	0.0730	0.00907	8.04	8.88e-16
choice_asc_pt	-10.8	4.17	-2.60	0.00938
choice_beta_time_pt	-12.6	2.78	-4.55	5.39e-06
choice_asc_car	24.1	6.52	3.70	0.000211
choice_beta_time_car	-27.3	4.71	-5.80	6.78e-09
choice_beta_car_centric_ attitude_car	9.79	1.95	5.01	5.34e-07
choice_beta_dist_work	-2.78	0.409	-6.81	9.62e-12
choice_beta_dist_other_ purposes	-4.44	0.720	-6.16	7.18e-10

Table 11: Estimation results for the simultaneous hybrid choice model (one latent variable, continuous measurement): choice model

measurement_coefficient_car_ centric_attitude_LifSty07	0.0626	0.0473	1.32	0.186
measurement_LifSty07_sigma_log	0.218	0.0251	8.69	0.00

Table 10: Estimation results for the simultaneous hybrid choice model (one latent variable, continuous measurement): measurement equations

The estimation results reported in Tables 9, 10, and 11 are broadly consistent with those obtained in the previous examples. The estimates of the structural equation and of the measurement equations remain remarkably stable relative to the MIMIC model of Section 4.2. In particular, the effects of car-oriented parents and high education on the car-centric attitude retain similar magnitudes and levels of significance, and the measurement loadings exhibit the same signs and interpretation as in Table 5. The most notable changes appear in the choice model. The coefficients associated with travel time and distance remain close to those obtained in the baseline and sequential specifications, confirming the robustness of the effects of the observed travel attributes. However, the coefficient associated with the car-centric latent variable increases substantially, from 3.80 in the sequential model (Table 7) to 9.79 in the simultaneous model (Table 11), and becomes highly significant. This result indicates that, once the measurement and choice components are estimated jointly, the latent attitude plays a much more important role in explaining mode choice. The simultaneous estimation therefore reveals a stronger relationship between attitudes and behavior than the sequential approach, illustrating one of the main motivations for estimating the full hybrid choice model.

4.5 Example 5: hybrid mode choice model with ordered-logit measurement

In this example, the hybrid choice model of Example 4.4 is kept unchanged on the choice side and on the latent-variable side, but the measurement model is modified in order to reflect the ordinal nature of the Likert indicators. The structural equation of the latent variable remains the one introduced previously, and the mode choice model is still specified through the hybrid likelihood of Eqs. (19)–(21). The only change concerns the measurement equations entering the term $\Lambda_n^I(I_n | \mathbf{x}_n^*, \mathbf{x}_n)$.

More precisely, the continuous measurement equations used in Example 4.4 are replaced by ordinal measurement equations of the type introduced in Section 1.3. For each indicator ℓ , we introduce an underlying latent response and a system of thresholds as in Eqs. (8)–(10). In the present example, the latent response is modeled as a linear function of the car-centric latent

Number of estimated parameters	41
Sample size	889
Init log likelihood	-20187.62
Final log likelihood	-10505.44
Akaike Information Criterion	21092.88
Bayesian Information Criterion	21289.27
Number of draws	50000
Optimization time	3:02:20.675567

Table 12: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered logit): summary statistics

variable, exactly as in Eq. (11). The contribution of each indicator to the conditional likelihood therefore has the same structure as Eq. (12), except that the normal cumulative distribution is replaced by the logistic cumulative distribution. The overall conditional indicator likelihood is then obtained, as in Eq. (16), by multiplying the contributions of all indicators.

The hybrid model itself remains identical in structure to the one introduced in Example 4.4. In particular, the latent variable still enters the utility of the car alternative, while the other utilities are unchanged. The full likelihood is therefore still given by the hybrid formulation of Eq. (20), with the sole difference that the measurement component now corresponds to an ordered-logit specification rather than to the continuous case.

The main conceptual novelty of this example is therefore not the hybrid structure, but the treatment of the indicators. In the previous example, the Likert responses were approximated as continuous variables. Here, they are modeled explicitly as ordinal outcomes, which is more natural for attitudinal statements recorded on ordered response scales. This requires estimating a threshold structure in addition to the intercepts, loadings, and measurement scales.

The normalization strategy also differs from that of the continuous case. As explained in Section 2, identification requires normalizations both at the latent-variable level and at the ordinal measurement level. The latent variable is normalized exactly as before using the reference-indicator strategy: for indicator `Envir01`, the intercept is fixed to zero and the loading on the latent variable is fixed to -1 . In addition, the scale of the ordinal measurement layer is anchored by fixing the measurement scale parameter of the same reference indicator to one. Consequently, all parameters associated with the measurement equation of `Envir01` are fixed and none of them is estimated. The reference indicator therefore serves exclusively to define the location, orientation, and scale of the latent-variable measurement system.

With these constraints, the remaining intercepts, loadings, threshold parameters, and measurement scales are interpreted relative to the chosen reference indicator and to the common ordinal measurement scale.

This specification is implemented in the script `plot_h05_mode_lv_ordlogit_simult.py`. Relative to the previous example, the model structure is unchanged; only the measurement equations and the associated normalization are modified in order to move from the continuous case to an explicit ordinal treatment of the indicators.

Description	Coeff. estimate	BHHH Asympt.		t-stat	p-value
		std.	error		
measurement_intercept_Envir02	1.07	0.154		6.94	4.01e-12
measurement_coefficient_car_ centric_attitude_Envir02	-0.525	0.0379		-13.8	0.00
measurement_Envir02_sigma_log	-0.164	0.110		-1.50	0.134
measurement_intercept_Envir06	2.44	0.244		10.0	0.00
measurement_coefficient_car_ centric_attitude_Envir06	-0.362	0.0339		-10.7	0.00

measurement_Envir06_sigma_log	-0.431	0.103	-4.19	2.82e-05
measurement_intercept_Mobil03	0.454	0.103	4.42	1.00e-05
measurement_coefficient_car_	-0.175	0.0352	-4.97	6.64e-07
centric_attitude_Mobil03				
measurement_Mobil03_sigma_log	0.0237	0.105	0.225	0.822
measurement_intercept_Mobil05	0.760	0.144	5.29	1.25e-07
measurement_coefficient_car_	-0.380	0.0434	-8.74	0.00
centric_attitude_Mobil05				
measurement_Mobil05_sigma_log	0.150	0.111	1.35	0.179
measurement_intercept_Mobil08	-1.13	0.142	-7.95	2.00e-15
measurement_coefficient_car_	0.315	0.0423	7.46	8.90e-14
centric_attitude_Mobil08				
measurement_Mobil08_sigma_log	0.129	0.109	1.18	0.237
measurement_intercept_Mobil09	1.69	0.192	8.80	0.00
measurement_coefficient_car_	-0.355	0.0338	-10.5	0.00
centric_attitude_Mobil09				
measurement_Mobil09_sigma_log	-0.207	0.106	-1.96	0.0501
measurement_intercept_Mobil10	0.0872	0.155	0.563	0.573
measurement_coefficient_car_	0.667	0.0779	8.57	0.00
centric_attitude_Mobil10				
measurement_Mobil10_sigma_log	0.440	0.132	3.32	0.000885
measurement_intercept_LifSty07	-1.29	0.140	-9.21	0.00
measurement_coefficient_car_	0.139	0.0366	3.80	0.000146
centric_attitude_LifSty07				
measurement_LifSty07_sigma_log	0.147	0.113	1.30	0.194

Table 14: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered logit): measurement equations

The estimation results reported in Tables 13, 14, 15, and 16 confirm the qualitative findings of the previous hybrid specifications. In the structural equation, the same variables remain relevant: individuals with car-oriented parents have a significantly higher car-centric attitude, whereas individuals with high education have a significantly lower one. The estimated measurement coefficients keep the same signs as in the continuous measurement model, but the ordinal specification strengthens the role of the indicators: all loadings are now statistically significant, including `LifSty07`, which was weak in the continuous case. The threshold parameters are also significant, indicating that the ordered-logit measurement layer is well identified. On the choice side, the coefficients of travel time and distance remain close to those obtained in the baseline, sequential, and simultaneous continuous models, confirming the stability of the observed-attribute effects. The coefficient of the car-centric latent variable remains positive and highly significant, although its magnitude is smaller than in the continuous measurement specification. Thus, explicitly treating the Likert indicators as ordinal preserves the main behavioral conclusions while providing a more appropriate measurement model for the psychometric indicators.

Description	Coeff. estimate	BHHH		t-stat	p-value
		Asympt.	std. error		
struct_car_centric_attitude_ intercept	0.352	0.243		1.45	0.148
struct_car_centric_attitude_ top_manager	0.320	0.294		1.09	0.277
struct_car_centric_attitude_ car_oriented_parents	0.706	0.213		3.31	0.000931
struct_car_centric_attitude_ high_education	-1.05	0.269		-3.90	9.50e-05
struct_car_centric_attitude_ low_education	0.379	0.260		1.46	0.145
struct_car_centric_attitude_ used_to_go_to_school_by_car	0.182	1.19		0.153	0.878
struct_car_centric_attitude_ sigma_log	0.756	0.0914		8.27	2.22e-16

Table 13: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered logit): structural equation

Description	Coeff. estimate	BHHH		t-stat	p-value
		Asympt.	std. error		
likert_delta_0.log	-0.686	0.0965		-7.11	1.16e-12
likert_delta_1.log	0.513	0.0882		5.82	5.88e-09

Table 15: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered logit): measurement thresholds

Description	Coeff. estimate	BHHH		t-stat	p-value
		Asympt.	std. error		
choice_scale_parameter	0.0699	0.00908		7.70	1.40e-14
choice_asc_pt	-11.5	4.32		-2.66	0.00771
choice_beta_time_pt	-13.3	2.98		-4.47	7.83e-06
choice_asc_car	-4.34	3.61		-1.20	0.230
choice_beta_time_car	-28.6	5.08		-5.62	1.88e-08
choice_beta_car_centric_ attitude_car	5.30	1.11		4.77	1.88e-06
choice_beta_dist_work	-2.91	0.436		-6.67	2.51e-11
choice_beta_dist_other_ purposes	-4.63	0.762		-6.08	1.21e-09

Table 16: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered logit): choice model

4.6 Example 6: hybrid mode choice model with ordered-probit measurement

This specification is identical to the ordered-logit hybrid model presented in Example 4.5. The structural equation, the utility specification, the hybrid likelihood (Eq. (20)), and the normalization strategy remain unchanged.

The only difference lies in the measurement equations for the Likert indicators. Instead of using the logistic cumulative distribution, the ordinal probabilities are computed using the normal cumulative distribution, as defined in Eq. (12). This leads to an ordered-probit measurement model.

The implementation is provided in the script `plot_h06_mode_lv_ordprobit_simult.py`.

Description	Coeff. estimate	BHHH		t-stat	p-value
		Asympt. std.	error		
measurement_intercept_Envir02	0.838	0.207		4.04	5.26e-05
measurement_coefficient_car_ centric_attitude_Envir02	-0.475	0.0585		-8.11	4.44e-16
measurement_Envir02.sigma_log	-0.111	0.184		-0.600	0.548
measurement_intercept_Envir06	1.39	0.252		5.53	3.25e-08
measurement_coefficient_car_ centric_attitude_Envir06	-0.319	0.0385		-8.29	2.22e-16
measurement_Envir06.sigma_log	-0.478	0.165		-2.90	0.00371
measurement_intercept_Mobil03	1.02	0.250		4.07	4.72e-05
measurement_coefficient_car_ centric_attitude_Mobil03	-0.198	0.0508		-3.91	9.36e-05
measurement_Mobil03.sigma_log	0.320	0.196		1.64	0.101
measurement_intercept_Mobil05	1.43	0.411		3.49	0.000490
measurement_coefficient_car_ centric_attitude_Mobil05	-0.512	0.0997		-5.13	2.90e-07
measurement_Mobil05.sigma_log	0.570	0.226		2.51	0.0119
measurement_intercept_Mobil08	0.247	0.122		2.03	0.0421
measurement_coefficient_car_ centric_attitude_Mobil08	0.424	0.0816		5.20	2.04e-07
measurement_Mobil08.sigma_log	0.511	0.215		2.38	0.0174
measurement_intercept_Mobil09	1.26	0.260		4.84	1.33e-06
measurement_coefficient_car_ centric_attitude_Mobil09	-0.327	0.0447		-7.33	2.29e-13
measurement_Mobil09.sigma_log	-0.153	0.179		-0.853	0.394
measurement_intercept_Mobil10	4.34	2.50		1.74	0.0822
measurement_coefficient_car_ centric_attitude_Mobil10	1.90	0.948		2.00	0.0453
measurement_Mobil10.sigma_log	1.57	0.547		2.86	0.00419

Number of estimated parameters	41
Sample size	889
Init log likelihood	-21722.37
Final log likelihood	-10532.96
Akaike Information Criterion	21147.93
Bayesian Information Criterion	21344.32
Number of draws	50000
Optimization time	3:46:12.509410

Table 17: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered probit): summary statistics

measurement_intercept_LifSty07	-0.0495	0.0980	-0.505	0.614
measurement_coefficient_car_ centric_attitude_LifSty07	0.156	0.0505	3.08	0.00204
measurement_LifSty07_sigma_log	0.466	0.222	2.10	0.0355

Table 19: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered probit): measurement equations

The estimation results reported in Tables 18, 19, 20, and 21 are qualitatively very similar to those obtained for the ordered-logit specification of Section 4.5. The same explanatory variables remain significant in the structural equation, namely **car_oriented_parents** with a positive effect and **high_education** with a negative effect on the car-centric attitude. Likewise, the measurement equations display the same pattern of signs and significance as in the ordered-logit model, confirming the interpretation of the latent variable and the consistency of the measurement structure. On the choice side, the coefficients associated with travel time, distance, and the scale parameter are almost identical to those obtained previously, demonstrating that the choice component is largely insensitive to the specific distributional assumption adopted for the ordinal indicators. The coefficient of the car-centric latent variable remains positive and highly significant, with an estimated value lying between those obtained for the ordered logit and the continuous measurement specifications. Overall, the ordered-probit and ordered-logit models lead to essentially the same behavioral conclusions. This similarity is expected, as the logistic and normal cumulative distributions have comparable shapes, and it illustrates the well-known empirical robustness of ordinal models with respect to the choice between logit and probit formulations.

Description	Coeff. estimate	BHHH		t-stat	p-value
		Asympt.	std. error		
struct_car_centric_attitude_ intercept	-0.270	0.156		-1.73	0.0829
struct_car_centric_attitude_ top_manager	0.154	0.183		0.841	0.401
struct_car_centric_attitude_ car_oriented_parents	0.441	0.132		3.33	0.000858
struct_car_centric_attitude_ high_education	-0.646	0.165		-3.91	9.26e-05
struct_car_centric_attitude_ low_education	0.245	0.161		1.52	0.128
struct_car_centric_attitude_ used_to_go_to_school_by_car	0.130	0.705		0.185	0.853
struct_car_centric_attitude_ sigma_log	0.293	0.0850		3.45	0.000563

Table 18: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered probit): structural equation

Description	Coeff. estimate	BHHH		t-stat	p-value
		Asympt.	std. error		
likert_delta_0_log	-2.48	0.273		-9.06	0.00
likert_delta_1_log	-0.278	0.152		-1.83	0.0668

Table 20: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered probit): measurement thresholds

Description	Coeff. estimate	BHHH		t-stat	p-value
		Asympt.	std. error		
choice_scale_parameter	0.0699	0.00907		7.71	1.29e-14
choice_asc_pt	-11.4	4.32		-2.64	0.00822
choice_beta_time_pt	-13.3	2.97		-4.47	7.83e-06
choice_asc_car	-0.245	3.67		-0.0669	0.947
choice_beta_time_car	-28.5	5.06		-5.62	1.87e-08
choice_beta_car_centric_ attitude_car	8.20	1.69		4.85	1.24e-06
choice_beta_dist_work	-2.90	0.436		-6.67	2.62e-11
choice_beta_dist_other_ purposes	-4.62	0.760		-6.07	1.24e-09

Table 21: Estimation results for the simultaneous hybrid choice model (one latent variable, ordered probit): choice model

5 Conclusion

This document has introduced the specification and estimation of hybrid choice models using Biogeme, combining structural equations for latent variables, measurement equations for indicators, and discrete choice models within a unified framework.

Hybrid choice models extend traditional discrete choice analysis by incorporating latent constructs that capture attitudes and perceptions. By linking these constructs simultaneously to observed indicators and to choices, they provide a richer and more behaviorally meaningful representation of decision-making processes.

A key aspect of these models is identification. Because latent variables and ordinal measurement systems are not directly observed, appropriate normalization is essential. The reference-indicator strategy, together with consistent normalization of ordinal thresholds, offers a practical and interpretable solution.

The case study has illustrated how these elements can be implemented step by step. In particular, the comparison between sequential and simultaneous estimation highlights the importance of accounting for the uncertainty of latent variables. The results show that simultaneous estimation yields more consistent and behaviorally plausible outcomes.

Finally, the transition from continuous to ordinal measurement models emphasizes the importance of aligning the specification with the nature of the data. While ordinal models are more demanding computationally, they provide a more appropriate representation of Likert-scale indicators.

For completeness, Appendix A provides a description of all variables used in the case study, including the explanatory variables entering the structural and choice models as well as the psychometric indicators. Appendix B contains the complete specification files corresponding to the examples presented throughout the document. These files illustrate how the theoretical concepts discussed in the main text are translated into executable Biogeme specifications and can serve as templates for the development of new hybrid choice models.

Overall, Biogeme offers a flexible and powerful environment for estimating hybrid choice models, enabling researchers to progressively build and analyze complex behavioral specifications.

A Description of the variables

The following table describes the variables involved in the models described in this document.

Name	Description
TimePT	The duration of the loop performed in public transport (in minutes).
WaitingTimePT	The total waiting time in a loop performed in public transports (in minutes).
TimeCar	The total duration of a loop made using the car (in minutes).
MarginalCostPT	The total cost of a loop performed in public transports, taking into account the ownership of a seasonal ticket by the respondent. If the respondent has a “GA” (full Swiss season ticket), a seasonal ticket for the line or the area, this variable takes value zero. If the respondent has a half-fare travelcard, this variable corresponds to half the cost of the trip by public transport..
CostCarCHF	The total gas cost of a loop performed with the car in CHF.

TripPurpose	The main purpose of the loop: 1 =Work-related trips; 2 =Work- and leisure-related trips; 3 =Leisure related trips. -1 represents missing values
UrbRur	Binary variable, where: 1 =Rural; 2 =Urban.
distance_km	Total distance performed for the loop.
age	Age of the respondent (in years) -1 represents missing values.
ResidChild	Main place of residence as a kid (< 18), 1 is city center (large town), 2 is city center (small town), 3 is suburbs, 4 is suburban town, 5 is country side (village), 6 is countryside (isolated), -1 is for missing data and -2 if respondent didn't answer to any opinion questions.
NbCar	Number of cars in the household.-1 for missing value.
NbBicy	Number of bikes in the household. -1 for missing value.
HouseType	House type, 1 is individual house (or terraced house), 2 is apartment (and other types of multi-family residential), 3 is independent room (subletting). -1 for missing value.
Income	Net monthly income of the household in CHF. 1 is less than 2500, 2 is from 2501 to 4000, 3 is from 4001 to 6000, 4 is from 6001 to 8000, 5 is from 8001 to 10'000 and 6 is more than 10'001. -1 for missing value.
CalculatedIncome	Net monthly income of the household in CHF, calculated as a continuous variable. The value is the center of the interval of the corresponding income category.
FamilSitu	Familiar situation: 1 is single, 2 is in a couple without children, 3 is in a couple with children, 4 is single with your own children, 5 is in a colocation, 6 is with your parents and 7 is for other situations. -1 for missing values.
SocioProfCat	To which of the following socioprofessional categories do you belong? 1 is for top managers, 2 for intellectual professions, 3 for freelancers, 4 for intermediate professions, 5 for artisans and salespersons, 6 for employees, 7 for workers and 8 for others. -1 for missing values.
GenAbST	Is equal to 1 if the respondent has a GA (full Swiss season ticket) and 2 if not.

Education	Highest education achieved. As mentioned by Wikipedia in English: "The education system in Switzerland is very diverse, because the constitution of Switzerland delegates the authority for the school system mainly to the cantons. The Swiss constitution sets the foundations, namely that primary school is obligatory for every child and is free in public schools and that the confederation can run or support universities." (source: Education in Switzerland (Wikipedia), accessed April 16, 2013). It is thus difficult to translate the survey that was originally in French and German. The possible answers in the survey are: 1. Unfinished compulsory education: education is compulsory in Switzerland but pupils may finish it at the legal age without succeeding the final exam. 2. Compulsory education with diploma. 3. Vocational education: a three or four-year period of training both in a company and following theoretical courses. Ends with a diploma called "Certificat fédéral de capacité" (i.e., "professional baccalaureate") (reference: Certificat fédéral de capacité (Wikipedia) - in French). 4. A 3-year generalist school giving access to teaching school, nursing schools, social work school, universities of applied sciences or vocational education (some-time in less than the normal number of years). It does not give access to universities in Switzerland. 5. High school: ends with the general baccalaureate exam. The general baccalaureate gives access automatically to universities. 6. Universities of applied sciences, teaching schools, nursing schools, social work schools: ends with a Bachelor and sometimes a Master, mostly focus on vocational training. 7. Universities and institutes of technology: ends with an academic Bachelor and in most cases an academic Master. 8. PhD thesis.
-----------	---

Table 22: Description of variables

B Complete specification files

This section presents the Python implementation of the hybrid choice model used in the case study. The following specification files have been used for the estimation of the results presented in this chapter. They have been developed and tested with Biogeme 3.3.3. It is possible that minor adaptations of the syntax may be required for future versions of Biogeme.

B.1 choice_latent_variables .py

```
1  """Choice model specification used in the latent-variable tutorial.
2
3  This module defines the systematic utility functions of the mode-choice
4  model. It illustrates how a choice model is specified independently of the
5  latent-variable model and how latent variables can subsequently be injected
6  into the utility functions.
7
8  The specification considers three transportation alternatives:
9
10 - public transport,
11 - car,
12 - slow modes.
13
14 The utilities are generated by the function
15 'generate_utility_functions' and returned as a dictionary indexed by
16 alternative identifiers. The resulting dictionary can be passed directly
17 to a Biogeme logit model.
18
19 The example is intentionally structured in two layers:
20
21 1. A baseline specification based exclusively on observed attributes
22   (travel time, travel cost, distance, and trip purpose).
23 2. An extended specification in which latent variables may enter the
24   utility functions through interaction terms.
25
26 The latent variables themselves are not defined in this module. They are
27 constructed elsewhere from structural and measurement equations and are
28 provided here as Biogeme expressions through the argument
29 'latent_expressions'. This separation reflects the conceptual structure
30 of hybrid choice models, where latent variables are first generated and
31 then incorporated into the choice model.
32
33 In the current example, the latent variable
34 'car_centric_attitude' may influence the utility of the car
35 alternative. If no latent expressions are provided, the function reduces
36 to the baseline mode-choice specification.
37
38 The cost coefficient is normalized to -1 and a positive scale parameter is
39 estimated. This normalization defines the unit of utility while allowing
40 the overall scale of the choice model to be estimated.
41
42 Michel Bierlaire
43 Fri Jun 12 2026, 15:00:22
44 """
45
46 from optima import (
47     CostCarCHF,
48     MarginalCostPT,
49     PurpWH,
50     TimeCar_hour,
51     TimePT_hour,
52     distance_km,
53 )
54
55 from biogeme.expressions import (
56     Beta,
57     Expression,
58 )
59
60
61 def generate_utility_functions(
62     latent_expressions: dict[str, Expression] | None = None,
63 ) -> dict[int, Expression]:
64     """Generate the systematic utility functions of the choice model.
65
66     The function creates the utility expressions associated with the three
67     transportation alternatives considered in the tutorial:
68
69     - alternative 0: public transport,
70     - alternative 1: car,
71     - alternative 2: slow modes.
72
73     The utilities are expressed using Biogeme expressions and include the
74     estimated parameters declared in this function.
75
76     Baseline specification
77     -----
78
79     Public transport:
80
81     .. math::
82
83         V_{\mathrm{PT}}
84         = \beta_{\mathrm{asc,PT}}
85         + \beta_{\mathrm{time,PT}} \backslash \mathrm{TimePT}
86         + \beta_{\mathrm{cost}} \backslash \mathrm{MarginalCostPT}
87
88     Car:
89
90     .. math::
```

```

91     V_{\mathrm{Car}}
92     = \beta_{\mathrm{asc,Car}}
93     + \beta_{\mathrm{time,Car}}\backslash,\mathrm{TimeCar}
94     + \beta_{\mathrm{cost}}\backslash,\mathrm{CostCar}
95
96
97 Slow modes:
98
99 .. math::
100
101     V_{\mathrm{Slow}}
102     = \beta_{\mathrm{dist}}\backslash,\mathrm{Distance}
103
104 where the distance coefficient depends on trip purpose.
105
106 Latent-variable extension
107 -----
108
109 When ‘‘latent_expressions’’ is provided, the latent variable
110 ‘‘car_centric_attitude’’ enters the utility of the car alternative
111 through an additional coefficient:
112
113 .. math::
114
115     V_{\mathrm{Car}}
116     \rightarrow
117     V_{\mathrm{Car}}
118     + \beta_{\mathrm{car\_centric}}
119     x_{\mathrm{car\_centric}}
120
121 This illustrates the standard mechanism used in hybrid choice models:
122 latent variables are treated as additional explanatory variables in the
123 utility specification.
124
125 Utility scaling
126 -----
127
128 After the utilities are constructed, they are multiplied by a positive
129 scale parameter. Because the cost coefficient is fixed to -1, the scale
130 parameter can be estimated and captures the overall sensitivity of
131 choice probabilities to utility differences.
132
133 :param latent_expressions: Optional dictionary mapping latent-variable
134     names to Biogeme expressions. If ‘‘None’’, the baseline choice model
135     is generated. The current specification expects the key
136     ‘‘car_centric_attitude’’ when latent variables are used.
137 :return: Dictionary mapping alternative identifiers to systematic
138     utility expressions.
139 """
140 work_trip = PurpHWH == 1
141 other_trip_purposes = PurpHWH != 1
142
143 # Choice model: parameters
144 choice_beta_cost = Beta('choice_beta_cost', -1, None, 0, 1)
145
146 choice_asc_car = Beta('choice_asc_car', 0.0, None, None, 0)
147
148 choice_asc_pt = Beta('choice_asc_pt', 0, None, None, 0)
149
150 choice_beta_dist_work = Beta('choice_beta_dist_work', 0, None, 0, 0)
151 choice_beta_dist_other_purposes = Beta(
152     'choice_beta_dist_other_purposes', 0, None, 0, 0
153 )
154 choice_beta_dist = (
155     choice_beta_dist_work * work_trip
156     + choice_beta_dist_other_purposes * other_trip_purposes
157 )
158
159 scale_parameter = Beta('choice_scale_parameter', 1, 0.0001, None, 0)
160
161 # Time coefficients with optional LV interactions
162 choice_beta_time_car = Beta('choice_beta_time_car', 0, None, 0, 0)
163 choice_beta_time_pt = Beta('choice_beta_time_pt', 0, None, 0, 0)
164
165 choice_beta_car_centric_attitude_car = Beta(
166     'choice_beta_car_centric_attitude_car', 0, None, None, 0
167 )
168 v_public_transport_det = (
169     choice_asc_pt
170     + choice_beta_time_pt * TimePT_hour
171     + choice_beta_cost * MarginalCostPT
172 )
173 v_public_transport = v_public_transport_det
174
175 v_car_det = (
176     choice_asc_car
177     + choice_beta_time_car * TimeCar_hour
178     + choice_beta_cost * CostCarCHF
179 )
180 v_car = (
181     v_car_det
182     if latent_expressions is None
183     else v_car_det
184     + choice_beta_car_centric_attitude_car
185     * latent_expressions['car_centric_attitude']
186 )
187
188 v_slow_modes_det = choice_beta_dist * distance_km
189 v_slow_modes = v_slow_modes_det
190
191 v = {
192     0: scale_parameter * v_public_transport,
193     1: scale_parameter * v_car,
194     2: scale_parameter * v_slow_modes,
195 }
196 return v

```

The file `choice_latent_variables.py` defines the utility specification used by the mode-choice component of the examples. Its main role is to keep the choice model separated from the latent-variable model. The function `generate_utility_functions` returns the systematic utilities of the three alternatives: public transport, car, and slow modes.

The baseline specification uses only observed travel attributes: travel time and marginal cost for public transport, travel time and cost for car, and distance for slow modes. The distance coefficient is allowed to depend on the trip purpose, distinguishing work trips from other trips.

The module also prepares the hybrid extension of the choice model. If a dictionary of latent-variable expressions is provided, the car-centric latent variable is added to the utility of the car alternative. If no such dictionary is provided, the same function generates the baseline logit model without latent variables. This design makes it possible to use the same utility specification in the standard choice model, in the sequential hybrid model, and in the simultaneous hybrid models.

The cost coefficient is fixed to -1 , which defines the utility unit. A positive scale parameter is then estimated and multiplies all systematic utilities. This reflects the normalization strategy discussed earlier: the cost normalization fixes the scale of utility, while the estimated scale parameter controls the sensitivity of the choice probabilities to utility differences.

B.2 `likert_spec.py`

```

1  """
2  Likert indicators and Likert types (SPEC ONLY)
3
4  Pure semantic indicator declarations
5  Michel Bierlaire
6  Sat Mar 14 2026, 16:26:11
7  """
8
9  from biogeme.latent_variables import LikertIndicator, LikertType
10
11  likert_indicators = [
12      LikertIndicator(
13          name="Envir01",
14          statement="Fuel price should be increased to reduce congestion and air pollution.",
15          type_name="likert",
16      ),
17      LikertIndicator(
18          name="Envir02",
19          statement="More public transportation is needed, even if taxes are set to pay the additional costs.",
20          type_name="likert",
21      ),
22      LikertIndicator(
23          name="Envir03",
24          statement="Ecology disadvantages minorities and small businesses.",
25          type_name="likert",
26      ),
27      LikertIndicator(
28          name="Envir04",
29          statement="People and employment are more important than the environment.",
30          type_name="likert",
31      ),
32      LikertIndicator(
33          name="Envir05",
34          statement="I am concerned about global warming.",
35          type_name="likert",
36      ),
37      LikertIndicator(
38          name="Envir06",
39          statement="Actions and decision making are needed to limit greenhouse gas emissions.",
40          type_name="likert",
41      ),
42      LikertIndicator(
43          name="Mobil03",
44          statement="I use the time of my trip in a productive way.",
45          type_name="likert",
46      ),
47      LikertIndicator(
48          name="Mobil05",
49          statement="I reconsider frequently my mode choice.",
50          type_name="likert",
51      ),
52      LikertIndicator(
53          name="Mobil08",
54          statement="I do not feel comfortable when I travel close to people I do not know.",
55          type_name="likert",
56      ),
57      LikertIndicator(
58          name="Mobil09",
59          statement="Taking the bus helps making the city more comfortable and welcoming.",
60          type_name="likert",
61      ),
62      LikertIndicator(
63          name="Mobil10",
64          statement="It is difficult to take the public transport when I travel with my children.",
65          type_name="likert",
66      ),
67      LikertIndicator(
68          name="Mobil12",
69          statement="It is very important to have a beautiful car.",

```

```

70         type_name="likert",
71     ),
72     LikertIndicator(
73         name="LifSty01",
74         statement="I always choose the best products regardless of price.",
75         type_name="likert",
76     ),
77     LikertIndicator(
78         name="LifSty07",
79         statement="The pleasure of having something beautiful consists in showing it.",
80         type_name="likert",
81     ),
82 ]
83
84 likert_types = [
85     LikertType(
86         type_name="likert",
87         symmetric=True,
88         categories=[1, 2, 3, 4, 5],
89         neutral_labels=[6, -1],
90     ),
91 ]

```

The file `likert_spec.py` contains the semantic declaration of the Likert indicators used in the measurement equations. Each indicator is represented by a `LikertIndicator` object, which associates an internal variable name, such as `Envir01` or `Mobil10`, with the corresponding survey statement. These declarations do not yet specify a statistical measurement equation. They only define the set of observed psychometric indicators that can later be linked to the latent variable.

The file also defines the Likert response scale through a `LikertType` object. In this example, all indicators share the same five-point symmetric scale, with response categories 1 to 5. The labels 6 and -1 are declared as neutral or non-informative responses, so that they can be treated separately when constructing the measurement likelihood. This separation between indicator declarations and the statistical measurement model reflects the semantic organization used throughout the examples: the present file defines what is observed, while the subsequent model scripts specify how these observations enter the likelihood.

B.3 number_of_draws.py

```

1  NUMBER_OF_DRAWS = 50_000

```

The file `number_of_draws.py` centralizes the number of draws used in the Monte-Carlo integration. It defines a single constant, `NUMBER_OF_DRAWS`, set to 50 000. This value is used by the examples involving latent variables, where the likelihood contains an integral over the distribution of the latent construct. Defining the number of draws in a separate file makes the examples easier to maintain and ensures that all models relying on simulation use the same level of numerical accuracy.

B.4 one_latent_variable_spec.py

```

1  """
2  Latent variables (SPEC ONLY)
3
4  Pure specification: what depends on what.
5
6  Michel Bierlaire
7  :Wed Mar 04 2026, 13:48:57
8  """
9
10 from biogeme.latent_variables import (
11     LatentVariable,
12     PositiveParameterSpec,
13     StructuralEquation,
14 )
15
16 DEFAULT_SIGMA_START = 10.0
17
18 car_centric_attitude = LatentVariable(
19     name='car_centric_attitude',
20     structural_equation=StructuralEquation(
21         name='car_centric_attitude',
22         intercept=True,
23         explanatory_variables=[
24             'top_manager',
25             'car_oriented_parents',
26             'high_education',
27             'low_education',
28             'used_to_go_to_school_by_car',
29         ],
30     ),
31     indicators={
32         'Envir01',
33         'Envir02',
34         'Envir06',
35         'Mobil03',

```

```

36         'Mobil05',
37         'Mobil08',
38         'Mobil09',
39         'Mobil10',
40         'LifSty07',
41     },
42     structural_sigma=PositiveParameterSpec(start=DEFAULT_SIGMA_START),
43 )
44
45 latent_variables = [car_centric_attitude]

```

This file defines the semantic specification of the latent variable used throughout the examples. It does not build a likelihood function and does not estimate any model. Instead, it describes the structure of the latent construct independently of the measurement and choice models in which it will later be used.

The object `car_centric_attitude` is declared as a `LatentVariable`. Its structural equation is specified by a `StructuralEquation` object with an intercept and five observed explanatory variables: `top_manager`, `car_oriented_parents`, `high_education`, `low_education`, and `used_to_go_to_school_by_car`. These variables correspond to the causes of the latent attitude in the MIMIC terminology introduced earlier.

The file also lists the psychometric indicators associated with the latent variable. These indicators are not modeled here; only their names are attached to the semantic specification. The actual measurement equations are introduced later, depending on whether the indicators are treated as continuous, ordered logit, or ordered probit responses.

Finally, the standard deviation of the structural disturbance is declared using a `PositiveParameterSpec`, ensuring that the corresponding parameter is constrained to be positive during estimation. The file ends by collecting the latent variable in the list `latent_variables`, which provides a common interface for the scripts that construct the different MIMIC and hybrid choice models.

B.5 optima.py

```

1  """
2  .. _optima_data:
3
4  Data preparation
5  =====
6
7  Optima data preparation for Biogeme.
8
9  This module reads the Optima case-study dataset from the tab-separated file
10 'optima.dat' and prepares a :class:'biogeme.database.Database' object.
11
12 The preparation consists of:
13
14 - **Filtering / cleaning**: remove observations that are incompatible with the
15   modeling assumptions (invalid choices, non-workers, tours with a single trip,
16   and observations with zero travel times or distance).
17 - **Feature engineering**: create derived variables used later in the choice and
18   latent-variable specifications (e.g., normalized weights, categories, scaled
19   variables).
20 - **Convenience exports**: expose a large set of
21   :class:'biogeme.expressions.Variable' objects corresponding to columns in the
22   dataset so that other modules can simply import them from here.
23
24 Michel Bierlaire
25 2023-04-12
26 """
27
28 import biogeme.database as db
29 import pandas as pd
30 from biogeme.expressions import Variable
31
32 data_file_path = 'optima.dat'
33
34
35 # %%
36 # Read the data
37 def read_data() -> db.Database:
38     """Read the data from file"""
39     df = pd.read_csv(data_file_path, sep='\t')
40     # Exclude observations such that the chosen alternative is -1
41     df.drop(df[df['Choice'] == -1].index, inplace=True)
42     # Exclude non workers
43     df = df[df['OccupStat'].isin([1, 2])]
44     # Exclude tours with 1 trip
45     df.drop(df[df['NbTrajects'] == 1].index, inplace=True)
46     # Exclude tours longer than 100km
47     df.drop(df[df['distance_km'] > 100].index, inplace=True)
48     # Exclude zero travel time
49     df.drop(df[df['TimePT'] == 0].index, inplace=True)
50     df.drop(df[df['TimeCar'] == 0].index, inplace=True)
51     df.drop(df[df['distance_km'] == 0].index, inplace=True)
52
53     car_not_available = df['CarAvail'] == 3
54     car_is_chosen = df['Choice'] == 1
55     incompatible = car_is_chosen & car_not_available
56     df.drop(df[incompatible].index, inplace=True)
57

```



```

58 df['worker'] = df['OccupStat'].isin([1, 2]).astype(int)
59 df['car_is_available'] = df['CarAvail'] != 3
60 # Normalize the weights
61 sum_weight = df['Weight'].sum()
62 number_of_rows = df.shape[0]
63 df['normalized_weight'] = df['Weight'] * number_of_rows / sum_weight
64 # Group car ownership: 3, 4, and 6 cars are grouped as 3
65 df['number_of_cars'] = df['NbCar'].replace({3: 3, 4: 3, 6: 3})
66 df.drop(df[df["number_of_cars"] == -1].index, inplace=True)
67
68 database = db.Database(name=data_file_path, dataframe=df)
69 - = database.define_variable('livesInUrbanArea', UrbRur == 2)
70 - = database.define_variable('owningHouse', OwnHouse == 1)
71 - = database.define_variable('used_to_go_to_school_by_car', ModeToSchool == 1)
72 - = database.define_variable('city_center_as_kid', ResidChild == 1)
73 - = database.define_variable('ScaledIncome', CalculatedIncome / 1000)
74 - = database.define_variable('age_65_more', age >= 65)
75 - = database.define_variable('age_30_less', age <= 30)
76 - = database.define_variable('age_category', 2 - (age <= 30) + (age >= 65))
77 - = database.define_variable(
78     'household_size', 3 - 2 * (NbHousehold == 1) - (NbHousehold == 2)
79 )
80 # 15% / 50% / 85 % quantiles of the income distribution in the data
81 - = database.define_variable(
82     'income_category',
83     1
84     + (CalculatedIncome >= 3250)
85     + (CalculatedIncome >= 7000)
86     + (CalculatedIncome >= 15000),
87 )
88 # % 30% / 60% quantile of the distance in the data
89 - = database.define_variable('distance_category', 1 + (distance_km >= 50))
90
91 - = database.define_variable('moreThanOneCar', NbCar > 1)
92 - = database.define_variable('moreThanOneBike', NbBicy > 1)
93 - = database.define_variable('individualHouse', HouseType == 1)
94 - = database.define_variable('male', Gender == 1)
95 - = database.define_variable('single', ((FamilSitu == 1) + (FamilSitu == 4)) > 0)
96 - = database.define_variable(
97     'haveChildren', ((FamilSitu == 3) + (FamilSitu == 4)) > 0
98 )
99 - = database.define_variable('haveGA', GenAbST == 1)
100 - = database.define_variable('high_education', Education >= 6)
101 - = database.define_variable('low_education', Education <= 3)
102 - = database.define_variable('top_manager', SocioProfCat == 1)
103 - = database.define_variable('artisans', SocioProfCat == 5)
104 - = database.define_variable('employees', SocioProfCat == 6)
105 - = database.define_variable(
106     'childCenter', ((ResidChild == 1) + (ResidChild == 2)) > 0
107 )
108
109 - = database.define_variable(
110     'childSuburb', ((ResidChild == 3) + (ResidChild == 4)) > 0
111 )
112 - = database.define_variable('car_oriented_parents', FreqCarPar == 4)
113 - = database.define_variable('TimePT_scaled', TimePT / 200)
114 - = database.define_variable('TimePT_hour', TimePT / 60)
115 - = database.define_variable('TimeCar_scaled', TimeCar / 200)
116 - = database.define_variable('TimeCar_hour', TimeCar / 60)
117 - = database.define_variable('MarginalCostPT_scaled', MarginalCostPT / 10)
118 - = database.define_variable('CostCarCHF_scaled', CostCarCHF / 10)
119 - = database.define_variable('distance_km_scaled', distance_km / 5)
120 - = database.define_variable('PurpHWH', TripPurpose == 1)
121 - = database.define_variable('PurpOther', TripPurpose != 1)
122 - = database.define_variable(
123     'number_of_trips',
124     (NbTrajects == 1) + 2 * (NbTrajects == 2) + 3 * (NbTrajects >= 3),
125 )
126 # urbanization: 1 = urban, 2 = mixed, 3 = rural
127 - = database.define_variable(
128     'urbanization', 2 - 1 * (TypeCommune <= 3) + 1 * (TypeCommune >= 8)
129 )
130
131 return database
132
133
134 # %%
135 # Variables from the data
136 Choice = Variable('Choice')
137 TimePT = Variable('TimePT')
138 TimeCar = Variable('TimeCar')
139 MarginalCostPT = Variable('MarginalCostPT')
140 CostCarCHF = Variable('CostCarCHF')
141 distance_km = Variable('distance_km')
142 Gender = Variable('Gender')
143 OccupStat = Variable('OccupStat')
144 Weight = Variable('Weight')
145 ID = Variable('ID')
146 DestAct = Variable('DestAct')
147 NbTransf = Variable('NbTransf')
148 WalkingTimePT = Variable('WalkingTimePT')
149 WaitingTimePT = Variable('WaitingTimePT')
150 CostPT = Variable('CostPT')
151 CostCar = Variable('CostCar')
152 NbHousehold = Variable('NbHousehold')
153 NbChild = Variable('NbChild')
154 NbCar = Variable('NbCar')
155 number_of_cars = Variable('number_of_cars')
156 NbMoto = Variable('NbMoto')
157 NbBicy = Variable('NbBicy')
158 NbBicyChild = Variable('NbBicyChild')
159 NbComp = Variable('NbComp')
160 NbTV = Variable('NbTV')
161 Internet = Variable('Internet')
162 NewspaperSubs = Variable('NewspaperSubs')
163 NbCellPhones = Variable('NbCellPhones')
164 NbSmartPhone = Variable('NbSmartPhone')

```

```

165 HouseType = Variable('HouseType')
166 OwnHouse = Variable('OwnHouse')
167 NbRoomsHouse = Variable('NbRoomsHouse')
168 YearsInHouse = Variable('YearsInHouse')
169 Income = Variable('Income')
170 BirthYear = Variable('BirthYear')
171 Mothertongue = Variable('Mothertongue')
172 FamilSitu = Variable('FamilSitu')
173 SocioProfCat = Variable('SocioProfCat')
174 CalculatedIncome = Variable('CalculatedIncome')
175 Education = Variable('Education')
176 HalfFareST = Variable('HalfFareST')
177 LineRelST = Variable('LineRelST')
178 GenAbST = Variable('GenAbST')
179 AreaRelST = Variable('AreaRelST')
180 OtherST = Variable('OtherST')
181 urbanization = Variable('urbanization')
182 three_trips_or_more = Variable('three_trips_or_more')
183 CarAvail = Variable('CarAvail')
184 Envir01 = Variable('Envir01')
185 Envir02 = Variable('Envir02')
186 Envir03 = Variable('Envir03')
187 Envir04 = Variable('Envir04')
188 Envir05 = Variable('Envir05')
189 Envir06 = Variable('Envir06')
190 Mobil01 = Variable('Mobil01')
191 Mobil02 = Variable('Mobil02')
192 Mobil03 = Variable('Mobil03')
193 Mobil04 = Variable('Mobil04')
194 Mobil05 = Variable('Mobil05')
195 Mobil06 = Variable('Mobil06')
196 Mobil07 = Variable('Mobil07')
197 Mobil08 = Variable('Mobil08')
198 Mobil09 = Variable('Mobil09')
199 Mobil10 = Variable('Mobil10')
200 Mobil11 = Variable('Mobil11')
201 Mobil12 = Variable('Mobil12')
202 Mobil13 = Variable('Mobil13')
203 Mobil14 = Variable('Mobil14')
204 Mobil15 = Variable('Mobil15')
205 Mobil16 = Variable('Mobil16')
206 Mobil17 = Variable('Mobil17')
207 Mobil18 = Variable('Mobil18')
208 Mobil19 = Variable('Mobil19')
209 Mobil20 = Variable('Mobil20')
210 Mobil21 = Variable('Mobil21')
211 Mobil22 = Variable('Mobil22')
212 Mobil23 = Variable('Mobil23')
213 Mobil24 = Variable('Mobil24')
214 Mobil25 = Variable('Mobil25')
215 Mobil26 = Variable('Mobil26')
216 Mobil27 = Variable('Mobil27')
217 ResidCh01 = Variable('ResidCh01')
218 ResidCh02 = Variable('ResidCh02')
219 ResidCh03 = Variable('ResidCh03')
220 ResidCh04 = Variable('ResidCh04')
221 ResidCh05 = Variable('ResidCh05')
222 ResidCh06 = Variable('ResidCh06')
223 ResidCh07 = Variable('ResidCh07')
224 LifSty01 = Variable('LifSty01')
225 LifSty02 = Variable('LifSty02')
226 LifSty03 = Variable('LifSty03')
227 LifSty04 = Variable('LifSty04')
228 LifSty05 = Variable('LifSty05')
229 LifSty06 = Variable('LifSty06')
230 LifSty07 = Variable('LifSty07')
231 LifSty08 = Variable('LifSty08')
232 LifSty09 = Variable('LifSty09')
233 LifSty10 = Variable('LifSty10')
234 LifSty11 = Variable('LifSty11')
235 LifSty12 = Variable('LifSty12')
236 LifSty13 = Variable('LifSty13')
237 LifSty14 = Variable('LifSty14')
238 TripPurpose = Variable('TripPurpose')
239 TypeCommune = Variable('TypeCommune')
240 UrbRur = Variable('UrbRur')
241 LangCode = Variable('LangCode')
242 ClassifCodeLine = Variable('ClassifCodeLine')
243 frequency = Variable('frequency')
244 ResidChild = Variable('ResidChild')
245 NbTrajects = Variable('NbTrajects')
246 FreqCarPar = Variable('FreqCarPar')
247 FreqTrainPar = Variable('FreqTrainPar')
248 FreqOtherPar = Variable('FreqOtherPar')
249 FreqTripHouseh = Variable('FreqTripHouseh')
250 Region = Variable('Region')
251 InVehicleTime = Variable('InVehicleTime')
252 ModeToSchool = Variable('ModeToSchool')
253 ReportedDuration = Variable('ReportedDuration')
254 CoderegionCAR = Variable('CoderegionCAR')
255 age = Variable('age')
256 age_category = Variable('age_category')
257 normalized_weight = Variable('normalized_weight')
258
259 ScaledIncome = Variable('ScaledIncome')
260 age_65_more = Variable('age_65_more')
261 moreThanOneCar = Variable('moreThanOneCar')
262 moreThanOneBike = Variable('moreThanOneBike')
263 individualHouse = Variable('individualHouse')
264 male = Variable('male')
265 haveChildren = Variable('haveChildren')
266 haveGA = Variable('haveGA')
267 high_education = Variable('high_education')
268 low_education = Variable('low_education')
269 childCenter = Variable('childCenter')
270 childSuburb = Variable('childSuburb')
271 TimePT_scaled = Variable('TimePT_scaled')

```

```

272 TimePT_hour = Variable('TimePT_hour')
273 TimeCar_scaled = Variable('TimeCar_scaled')
274 TimeCar_hour = Variable('TimeCar_hour')
275 MarginalCostPT_scaled = Variable('MarginalCostPT_scaled')
276 CostCarCHF_scaled = Variable('CostCarCHF_scaled')
277 distance_km_scaled = Variable('distance_km_scaled')
278 PurpHWH = Variable('PurpHWH')
279 PurpOther = Variable('PurpOther')
280 livesInUrbanArea = Variable('livesInUrbanArea')
281 household_size = Variable('household_size')
282 income_category = Variable('income_category')
283 distance_category = Variable('distance_category')
284 worker = Variable('worker')
285 car_is_available = Variable('car_is_available')

```

The file `optima.py` is responsible for preparing the data used throughout the examples. It reads the Optima case-study data from the file `optima.dat`, applies a number of filters, creates derived variables, and returns a Biogeme Database object.

The function `read_data` first imports the data using `pandas`. Observations that are not compatible with the modeling assumptions are removed: invalid choices, non-workers, tours with only one trip, observations with zero travel times or zero distance, and observations where the car alternative is chosen despite being unavailable. This cleaning step defines the estimation sample used in the subsequent examples.

The function then creates several variables used in the choice and latent-variable specifications. These include availability indicators, normalized sampling weights, socio-demographic categories, education indicators, car-availability indicators, trip-purpose indicators, and scaled versions of travel time, cost, and distance. The scaling of continuous variables is purely numerical: it improves the conditioning of the optimization problem without changing the behavioral content of the model.

Finally, the lower part of the file defines a large collection of Biogeme Variable objects. These objects provide symbolic access to the columns of the database and to the variables created inside `read_data`. Other modules can therefore import the variables directly from `optima.py`, which keeps the model specifications concise and separates data preparation from the definition of utilities, structural equations, and measurement equations.

B.6 `plot_h01_mode_logit.py`

```

1  """
2  Baseline mode choice model: maximum likelihood estimation
3  =====
4
5  This example estimates a multinomial logit model for transportation mode
6  choice using maximum likelihood and Biogeme.
7
8  It is the baseline specification of the hybrid-choice tutorial. The model
9  contains only observed choice variables and standard utility functions: there
10 are no latent variables, structural equations, or measurement equations. The
11 results therefore provide a reference against which the hybrid-choice
12 specifications introduced later can be compared.
13
14 The script performs the following steps:
15
16 - load the Optima mode-choice data,
17 - build the utility functions from the common tutorial specification,
18 - construct the logit log-likelihood,
19 - estimate the model, or reload previously saved estimation results,
20 - display the estimated parameters as pandas and LaTeX tables.
21
22 Michel Bierlaire
23 Sat Jun 06 2026, 15:23:41
24 """
25
26 from choice_latent_variables import generate_utility_functions
27 from optima import (
28     Choice,
29     read_data,
30 )
31
32 import biogeme.biogeme.logging as blog
33 from biogeme.biogeme import BIOGEME
34 from biogeme.models import loglogit
35 from biogeme.results_processing import (
36     get_latex_estimated_parameters,
37     get_latex_general_statistics,
38     get_pandas_estimated_parameters,
39 )
40
41 logger = blog.get_screen_logger(level=blog.INFO)
42
43 # %%
44 # Load the Optima mode-choice data.
45 database = read_data()
46
47 # Build the utility functions for the transportation alternatives.
48 utilities = generate_utility_functions()
49

```

```

50 # Construct the log-likelihood of the multinomial logit model.
51 log_likelihood = loglogit(utilities, None, Choice)
52
53 # Create the Biogeme object used for estimation.
54 biogeme = BIOGEME(
55     database,
56     log_likelihood,
57 )
58 biogeme.model_name = 'plot_h01_mode_logit'
59
60 # Estimate the model, or reload the saved results if they are already available.
61 yaml_file_name = f'saved_results/{biogeme.model_name}.yaml'
62 results = biogeme.estimate_or_load(yaml_file_name=yaml_file_name)
63
64 # Display a compact summary and the estimated parameters.
65 print(results.short_summary())
66 print(get_pandas_estimated_parameters(estimation_results=results))
67
68 general_statistics = get_latex_general_statistics(estimation_results=results)
69 print(general_statistics)
70
71 estimated_parameters = get_latex_estimated_parameters(estimation_results=results)
72 print(estimated_parameters[''])

```

This script implements the baseline model introduced in Section 4.1. It contains only the choice component of the model: no latent variable, structural equation, or measurement equation is involved. The script first reads the Optima data set using the common data-loading function defined in `optima.py`. It then obtains the systematic utilities from `generate_utility_functions`, so that the utility specification is shared with the hybrid models introduced later. The log-likelihood is constructed with Biogeme’s `loglogit` function, using the observed choice variable `Choice`.

The resulting expression is passed to a BIOGEME object, which performs maximum-likelihood estimation. The call to `estimate_or_load` either estimates the model or reloads previously saved results from the corresponding YAML file. Finally, the script prints a short summary of the estimation results and generates both pandas and L^AT_EX tables of the estimated parameters. This output provides the reference results against which the latent-variable and hybrid-choice specifications are compared in the following examples.

B.7 `plot_h02_lv_mimic_gauss.py`

```

1  """
2  Gaussian MIMIC model: maximum likelihood estimation
3  =====
4
5  This example estimates a Gaussian MIMIC model with one latent variable and no
6  choice component. The observed Likert indicators are treated as continuous
7  responses and modeled with Gaussian measurement equations.
8
9  The example introduces the latent-variable-only setting used later in the
10 hybrid-choice tutorial. The latent-variable structure itself is imported from a
11 separate semantic specification file. This script adds the Gaussian measurement
12 configuration, the normalization constraints required for identification, and
13 the maximum-likelihood estimation setup.
14
15 The script performs the following steps:
16
17 - load the latent-variable and indicator specifications,
18 - define a Gaussian measurement configuration for all indicators,
19 - define the normalization constraints used for identification,
20 - resolve the semantic specification into an estimable model,
21 - build the Biogeme expressions from the resolved model,
22 - integrate over the latent variable using Monte Carlo draws,
23 - construct the corresponding log-likelihood,
24 - estimate the model, or reload previously saved estimation results,
25 - display the estimated parameters as pandas and LaTeX tables.
26
27 Michel Bierlaire
28 Mon Jun 15 2026, 19:55:05
29 """
30
31 from __future__ import annotations
32
33 from likert_spec import likert_indicators, likert_types
34 from number_of_draws import NUMBER_OF_DRAWNS
35 from one_latent_variable_spec import DEFAULT_SIGMA_START, latent_variables
36 from optima import read_data
37
38 import biogeme.biogeme.logging as blog
39 from biogeme.biogeme import BIOGEME
40 from biogeme.expressions import log
41 from biogeme.latent_variables import (
42     BuildContext,
43     EstimationMode,
44     Fixing,
45     IndicatorMeasurementSpec,
46     MeasurementConfiguration,
47     MeasurementIntercept,
48     MeasurementLoading,
49     MeasurementModel,
50     NormalizationPlan,
51     PositivityParameterSpec,
52     PositivityMode,

```

```

53     build_biogeme_model ,
54     resolve_model ,
55 )
56 from biogeme.results_processing import (
57     get_latex_estimated_parameters ,
58     get_latex_general_statistics ,
59     get_pandas_estimated_parameters ,
60 )
61
62 logger = blog.get_screen_logger( level=blog.INFO)
63
64 # %%
65 # Gaussian measurement configuration for all Likert indicators.
66 measurement_configuration = MeasurementConfiguration(
67     specifications=[
68         IndicatorMeasurementSpec(
69             indicator_name=indicator.name,
70             measurement_model=MeasurementModel.GAUSSIAN,
71             measurement_sigma=PositiveParameterSpec( start=DEFAULT_SIGMA_START ) ,
72         )
73         for indicator in likert_indicators
74     ]
75 )
76
77 # %%
78 # Load the Optima data.
79 # -----
80 database = read_data()
81
82 # %%
83 # Define the build context for maximum likelihood estimation.
84 # -----
85 # The build context specifies how the semantic latent-variable specification is
86 # translated into estimable Biogeme expressions. We start from the default
87 # maximum-likelihood context and explicitly select the log-exp parameterization
88 # for positive parameters.
89 default_context = BuildContext.default( EstimationMode.MAXIMUM_LIKELIHOOD )
90 context = BuildContext(
91     estimation_mode=default_context.estimation_mode ,
92     draw_type=default_context.draw_type ,
93     positivity_mode=PositivityMode.LOG_EXP ,
94     naming=default_context.naming ,
95     ordinal_eps=default_context.ordinal_eps ,
96     ordinal_enforce_order=default_context.ordinal_enforce_order ,
97 )
98
99 # %%
100 # Identification constraints for the latent variable
101 # -----
102 # A latent variable is not directly observed, so its location and scale are
103 # not identified unless we impose normalization constraints.
104 #
105 # We use a reference-indicator strategy based on indicator 'Envir01':
106 #
107 # 1. The intercept of the reference indicator is fixed to 0.
108 #    This anchors the location of the latent-variable scale.
109 #
110 # 2. The loading of the reference indicator on the latent variable is fixed
111 #    to -1.
112 #    This anchors the scale of the latent variable and fixes its orientation.
113 #    The negative sign means that higher values of the latent variable imply
114 #    lower expected values for 'Envir01'.
115 #
116 # With these two constraints in place, the remaining intercepts and loadings
117 # are interpreted relative to the reference indicator.
118 normalization_plan = NormalizationPlan()
119
120 normalization_plan.add(
121     Fixing( MeasurementIntercept( 'Envir01' ), 0.0, note='reference indicator: location' )
122 )
123 normalization_plan.add(
124     Fixing(
125         MeasurementLoading( 'car_centric_attitude', 'Envir01' ),
126         -1.0,
127         note='reference indicator: scale and orientation',
128     )
129 )
130
131 # %%
132 # Resolve the semantic specification.
133 # -----
134 # The resolver combines the latent-variable specification, the indicator
135 # definitions, the Gaussian measurement configuration, and the normalization plan
136 # into an internal resolved model.
137 resolved_model = resolve_model(
138     latent_variables=latent_variables ,
139     likert_indicators=likert_indicators ,
140     likert_types=likert_types ,
141     measurement_configuration=measurement_configuration ,
142     context=context ,
143     normalization_plan=normalization_plan ,
144 )
145
146 # %%
147 # Build the Biogeme expressions.
148 # -----
149 # The builder translates the resolved model into expressions that can be used by
150 # Biogeme for maximum-likelihood estimation.
151 built_model = build_biogeme_model( resolved_model )
152
153 # %%
154 # Log-likelihood
155 # -----
156 # The builder provides the Monte-Carlo integrated measurement likelihood.
157 # Taking its logarithm yields the log-likelihood used for maximum-likelihood
158 # estimation.
159 integrated_likelihood = built_model.integrated_likelihood

```

```

160 log_likelihood = log(integrated_likelihood)
161
162 # %%
163 # Estimate the model with Biogeme.
164 #
165 # Existing results are reloaded from the YAML file when available.
166 biogeme = BIOGEME(
167     database,
168     log_likelihood,
169     number_of_draws=NUMBER_OF_DRAWS,
170     calculating_second_derivatives='never',
171     numerically_safe=False,
172     max_iterations=5_000,
173     group_of_parameters=built_model.parameter_groups,
174 )
175 biogeme.model_name = 'plot_h02_lv_mimic_gauss'
176
177 yaml_file_name = f'saved_results/{biogeme.model_name}.yaml'
178 results = biogeme.estimate_or_load(yaml_file_name=yaml_file_name)
179
180 # %%
181 # Display a compact summary and the estimated parameters.
182 print(results.short_summary())
183 pandas_results = get_pandas_estimated_parameters(
184     estimation_results=results,
185     group_of_parameters=built_model.parameter_groups,
186 )
187 for group_name, pandas_table in pandas_results.items():
188     print(group_name if group_name else 'Estimated parameters')
189     print(pandas_table)
190
191 general_statistics = get_latex_general_statistics(estimation_results=results)
192 print(general_statistics)
193
194 estimated_parameters = get_latex_estimated_parameters(
195     estimation_results=results,
196     group_of_parameters=built_model.parameter_groups,
197 )
198 for group_name, latex_table in estimated_parameters.items():
199     print(group_name if group_name else 'Estimated parameters')
200     print(latex_table)

```

The file `plot_h02_lv_mimic_gauss.py` implements the Gaussian MIMIC model discussed in Section 4.2. It is the first example where the latent-variable machinery is used explicitly, but it does not yet contain a choice component. The purpose of the script is therefore to estimate the structural equation of the latent variable together with continuous measurement equations for the Likert indicators.

The script first imports the semantic specification of the latent variable from `one_latent_variable_spec.py`, the list of Likert indicators from `likert_spec.py`, and the Optima data from `optima.py`. It then defines a `MeasurementConfiguration` in which each Likert indicator is associated with a Gaussian measurement model. In this specification, the indicators are treated as continuous responses, and each measurement equation has its own positive standard deviation parameter.

The next part of the script defines the build context used for maximum likelihood estimation. Positive parameters are represented through a log-exponential parametrization, which guarantees positivity while leaving the optimizer unconstrained in the transformed parameter. The normalization plan then imposes the identification constraints for the latent variable. The indicator `Envir01` is used as the reference indicator: its intercept is fixed to zero, and its loading on the car-centric latent variable is fixed to -1 . These two constraints anchor respectively the location, the scale, and the orientation of the latent-variable scale.

The semantic model is then resolved by combining the latent-variable specification, the indicator definitions, the Gaussian measurement configuration, and the normalization plan. The resolved model is translated into Biogeme expressions by `build_biogeme_model`. In particular, the builder provides the measurement likelihood integrated over the latent variable using Monte-Carlo simulation. Taking the logarithm of this integrated likelihood gives the log-likelihood maximized by Biogeme.

Finally, the script creates the BIOGEME object, specifies the number of simulation draws, estimates the model or reloads saved results, and prints the output both as pandas tables and as LaTeX tables. The parameter groups generated by the resolved model are used to organize the output according to the structural equation and the measurement equations.

B.8 plot_h03_mode_lv_gauss_seq.py

```

1 """
2 Sequential estimation of a choice model with a latent variable
3 =====
4
5 This example illustrates the second step of a sequential estimation

```

```

6 procedure.
7
8 The latent variable and its structural equation are first estimated using the
9 Gaussian MIMIC model of 'plot_h02_lv_mimic_gauss'. The estimated structural
10 parameters are then loaded from the saved results file and used to construct
11 the latent-variable expression appearing in the choice model.
12
13 Only the choice-model parameters are estimated in this script. The latent
14 variable is not re-estimated jointly with the measurement equations.
15
16 Michel Bierlaire
17 Mon Jun 15 2026, 19:55:16
18 ""
19
20 import sys
21
22 from choice_latent_variables import generate_utility_functions
23 from number_of_draws import NUMBER_OF_DRAWNS
24 from one_latent_variable_spec import car_centric_attitude
25 from optima import (
26     Choice,
27     read_data,
28 )
29
30 import biogeme.biogeme_logging as blog
31 from biogeme.biogeme import BIOGEME
32 from biogeme.expressions import (
33     Draws,
34     MonteCarlo,
35     MultipleSum,
36     Variable,
37     exp,
38     log,
39 )
40 from biogeme.models import logit
41 from biogeme.results_processing import (
42     EstimationResults,
43     get_latex_estimated_parameters,
44     get_latex_general_statistics,
45     get_pandas_estimated_parameters,
46 )
47
48 logger = blog.get_screen_logger(level=blog.INFO)
49
50 # %%
51 # Load data
52 database = read_data()
53
54 # %%
55 # Read the results of the previously estimated MIMIC model.
56 FILE_WITH_ESTIMATION_RESULTS = 'saved_results/plot_h02_lv_mimic_gauss.yaml'
57 try:
58     estimation_results = EstimationResults.from_yaml_file(
59         filename=FILE_WITH_ESTIMATION_RESULTS
60     )
61 except FileNotFoundError:
62     print(
63         f'File {FILE_WITH_ESTIMATION_RESULTS} does not exist. Sequential estimation cannot be performed.'
64     )
65     sys.exit()
66 estimated_parameters: dict[str, float] = estimation_results.get_beta_values()
67
68 # %%
69 # Reconstruct the latent-variable expression using the parameter estimates
70 # obtained from the MIMIC model. The sequential specification keeps both the
71 # deterministic component and the random structural disturbance.
72 list_of_explanatory_variables = (
73     car_centric_attitude.structural_equation.explanatory_variables
74 )
75
76 # The sigma parameter was estimated on the log scale in the MIMIC model.
77 # We therefore exponentiate it before scaling the structural disturbance.
78 terms = [
79     estimated_parameters[f'struct_car_centric_attitude_{var}'] * Variable(var)
80     for var in list_of_explanatory_variables
81 ]
82
83 structural_sigma = exp(estimated_parameters['struct_car_centric_attitude_sigma_log'])
84 structural_error = structural_sigma * Draws(
85     'car_centric_attitude_draw', 'NORMAL_MLHS_ANTI'
86 )
87
88 car_centric_attitude = MultipleSum(terms) + structural_error
89 latent_expressions = {'car_centric_attitude': car_centric_attitude}
90
91 # %%
92 # Utility functions including the latent-variable effect.
93 v = generate_utility_functions(latent_expressions=latent_expressions)
94
95 # %%
96 # Choice model estimated conditionally on the latent-variable specification.
97 conditional_likelihood = logit(v, None, Choice)
98
99 loglikelihood = log(MonteCarlo(conditional_likelihood))
100
101 biogeme = BIOGEME(
102     database,
103     loglikelihood,
104     number_of_draws=NUMBER_OF_DRAWNS,
105     numerically_safe=True,
106     max_iterations=5000,
107 )
108 biogeme.model_name = 'plot_h03_model_lv_gauss_seq'
109
110 yaml_file_name = f'saved_results/{biogeme.model_name}.yaml'
111 results = biogeme.estimate_or_load(yaml_file_name=yaml_file_name)
112

```



```

113 print(results.short_summary())
114 print(get_pandas_estimated_parameters(estimation_results=results))
115
116 general_statistics = get_latex_general_statistics(estimation_results=results)
117 print(general_statistics)
118
119 estimated_parameters = get_latex_estimated_parameters(estimation_results=results)
120 for group_name, latex_table in estimated_parameters.items():
121     print(group_name if group_name else 'Estimated parameters')
122     print(latex_table)

```

The script `plot_h03_mode_lv_gauss_seq.py` implements the sequential estimation approach discussed in Section 4.3. It assumes that the latent-variable model has already been estimated by `plot_h02_lv_mimic_gauss.py`, and starts by loading the corresponding estimation results from the saved YAML file. If this file is not available, the script stops, since the sequential estimation cannot be performed without the first-step estimates.

The next part of the script reconstructs the car-centric latent variable using the estimated structural-equation parameters. The explanatory variables are taken directly from the semantic specification of the latent variable, ensuring consistency with the MIMIC model. The estimated standard deviation of the structural disturbance is stored on the log scale and is therefore exponentiated before being multiplied by a normal draw. The resulting expression contains both the deterministic component of the structural equation and the random structural error term.

The reconstructed latent variable is then passed to `generate_utility_functions`, which inserts it into the utility of the car alternative. The choice probability is evaluated using the logit model, conditionally on the simulated value of the latent variable. Because the latent variable remains random, the conditional likelihood is integrated by Monte-Carlo simulation. The script then creates the BIOGEME object, estimates or loads the model, and prints the usual summary statistics and estimated parameters in both pandas and L^AT_EX formats.

This script therefore illustrates the main feature of sequential hybrid choice estimation: the measurement model is not part of the second-step likelihood. The indicators influence the choice model only through the latent-variable parameters estimated previously, which are treated as fixed in this script.

B.9 `plot_h04_mode_lv_gauss_simult.py`

```

1  """
2  Gaussian hybrid mode choice model: simultaneous maximum likelihood estimation
3  =====
4
5  This example estimates a hybrid mode choice model with one latent variable and
6  Gaussian measurement equations. The observed Likert indicators are treated as
7  continuous responses and modeled with Gaussian measurement equations.
8
9  In contrast with the sequential specification introduced previously, the
10 measurement component and the mode-choice component are estimated jointly. The
11 conditional likelihood of the indicators and the conditional likelihood of the
12 observed choice are multiplied conditionally on the latent variable, and the
13 resulting expression is integrated over the latent-variable distribution.
14
15 The latent-variable structure is imported from a separate semantic specification
16 file. This script adds the Gaussian measurement configuration, the normalization
17 constraints required for identification, the mode-choice utilities, and the
18 simultaneous maximum-likelihood estimation setup.
19
20 The script performs the following steps:
21
22 - load the latent-variable and indicator specifications,
23 - define a Gaussian measurement configuration for all indicators,
24 - define the normalization constraints used for identification,
25 - resolve the semantic specification into an estimable model,
26 - build the Biogeme expressions from the resolved model,
27 - build the choice utilities, including the latent-variable term,
28 - combine the measurement and choice conditional likelihoods,
29 - integrate the combined conditional likelihood over the latent variable,
30 - estimate the hybrid model, or reload previously saved estimation results,
31 - display the estimated parameters as grouped pandas and LaTeX tables.
32
33 Michel Bierlaire
34 Mon Jun 15 2026, 19:55:23
35 """
36
37 from __future__ import annotations
38
39 from choice_latent_variables import generate_utility_functions
40 from likert_spec import likert_indicators, likert_types
41 from number_of_draws import NUMBER_OF_DRAWNS
42 from one_latent_variable_spec import latent_variables
43 from optima import Choice, read_data
44
45 import biogeme.biogeme_logging as blog
46 from biogeme.biogeme import BIOGEME
47 from biogeme.expressions import MonteCarlo, log

```



```

48 from biogeme.latent_variables import (
49     BuildContext,
50     BuiltBiogemeModel,
51     EstimationMode,
52     Fixing,
53     IndicatorMeasurementSpec,
54     MeasurementConfiguration,
55     MeasurementIntercept,
56     MeasurementLoading,
57     MeasurementModel,
58     NormalizationPlan,
59     PositiveParameterSpec,
60     PositivityMode,
61     build_biogeme_model,
62     resolve_model,
63 )
64 from biogeme.models import logit
65 from biogeme.results_processing import (
66     get_latex_estimated_parameters,
67     get_latex_general_statistics,
68     get_pandas_estimated_parameters,
69 )
70
71 logger = blog.get_screen_logger(level=blog.INFO)
72
73 DEFAULT_MEASUREMENT_SIGMA_START = 10.0
74
75 # %%
76 # Gaussian measurement configuration for all Likert indicators.
77 # -----
78 measurement_configuration = MeasurementConfiguration(
79     specifications=[
80         IndicatorMeasurementSpec(
81             indicator_name=indicator.name,
82             measurement_model=MeasurementModel.GAUSSIAN,
83             measurement_sigma=PositiveParameterSpec(
84                 start=DEFAULT_MEASUREMENT_SIGMA_START
85             ),
86         )
87         for indicator in likert_indicators
88     ]
89 )
90
91 # %%
92 # Load the Optima data.
93 # -----
94 database = read_data()
95
96 # %%
97 # Define the build context for maximum likelihood estimation.
98 # -----
99 # The build context specifies how the semantic latent-variable specification is
100 # translated into estimable Biogeme expressions. We start from the default
101 # maximum-likelihood context and explicitly select the log-exp parameterization
102 # for positive parameters.
103 default_context = BuildContext.default(EstimationMode.MAXIMUM_LIKELIHOOD)
104 context = BuildContext(
105     estimation_mode=default_context.estimation_mode,
106     draw_type=default_context.draw_type,
107     positivity_mode=PositivityMode.LOG_EXP,
108     naming=default_context.naming,
109     ordinal_eps=default_context.ordinal_eps,
110     ordinal_enforce_order=default_context.ordinal_enforce_order,
111 )
112
113 # %%
114 # Identification constraints for the latent variable
115 # -----
116 # A latent variable is not directly observed, so its location and scale are
117 # not identified unless normalization constraints are imposed.
118 #
119 # We use a reference-indicator strategy based on 'Envir01':
120 # - the intercept of 'Envir01' is fixed to 0.0, which anchors the location
121 #   of the latent-variable scale;
122 # - the loading of 'Envir01' on 'car-centric-attitude' is fixed to -1.0,
123 #   which anchors the scale and fixes the orientation of the latent variable.
124 #
125 # The negative sign means that higher values of 'car-centric-attitude' imply
126 # lower expected values for 'Envir01'. With these constraints in place, the
127 # remaining intercepts and loadings are interpreted relative to the reference
128 # indicator.
129 normalization_plan = NormalizationPlan()
130
131 normalization_plan.add(
132     Fixing(
133         MeasurementIntercept('Envir01'),
134         0.0,
135         note='car-centric-attitude reference indicator: location',
136     )
137 )
138 normalization_plan.add(
139     Fixing(
140         MeasurementLoading('car-centric-attitude', 'Envir01'),
141         -1.0,
142         note='car-centric-attitude reference indicator: scale and orientation',
143     )
144 )
145
146 # %%
147 # Resolve the semantic specification.
148 # -----
149 # The resolver combines the latent-variable specification, the indicator
150 # definitions, the Gaussian measurement configuration, and the normalization plan
151 # into an internal resolved model.
152 resolved_model = resolve_model(
153     latent_variables=latent_variables,
154     likert_indicators=likert_indicators,

```

```

155     likert_types=likert_types ,
156     measurement_configuration=measurement_configuration ,
157     context=context ,
158     normalization_plan=normalization_plan ,
159 )
160
161 # %%
162 # Build the Biogeme expressions.
163 # -----
164 # The builder translates the resolved model into expressions that can be used by
165 # Biogeme for simultaneous maximum-likelihood estimation. It also provides
166 # report-ready parameter groups based on the parameters that are actually
167 # estimated.
168 built_model: BuiltBiogemeModel = build_biogeme_model(resolved_model)
169
170 # %%
171 # Choice utilities including the latent-variable term.
172 # -----
173 utilities = generate_utility_functions(built_model.latent_expressions)
174
175 # %%
176 # Conditional likelihood of the mode choice model
177 # -----
178 conditional_choice_likelihood = logit(utilities , None, Choice)
179
180 # %%
181 # Combined conditional likelihood
182 # -----
183 # The measurement component and the mode-choice component are combined
184 # conditionally on the latent variable. This is the key difference with the
185 # sequential approach: both parts are estimated jointly in a single likelihood.
186 combined_conditional_likelihood = (
187     built_model.conditional_likelihood * conditional_choice_likelihood
188 )
189
190 # %%
191 # Log-likelihood
192 # -----
193 # The combined conditional likelihood is integrated over the latent variable
194 # by Monte Carlo. Taking the logarithm of the resulting integrated likelihood
195 # yields the log-likelihood used for simultaneous maximum-likelihood
196 # estimation.
197 integrated_likelihood = MonteCarlo(combined_conditional_likelihood)
198 log_likelihood = log(integrated_likelihood)
199
200 # %%
201 # Estimate the model with Biogeme.
202 # -----
203 # Existing results are reloaded from the YAML file when available.
204 biogeme = BIOGEME(
205     database ,
206     log_likelihood ,
207     number_of_draws=NUMBER_OF_DRAWS,
208     calculating_second_derivatives='never' ,
209     max_iterations=5_000 ,
210     group_of_parameters=built_model.parameter_groups ,
211 )
212 biogeme.model_name = 'plot_h04_mode_lv_gauss_simult'
213
214 yaml_file_name = f'saved_results/{biogeme.model_name}.yaml'
215 results = biogeme.estimate_or_load(yaml_file_name=yaml_file_name)
216
217 # %%
218 # Display a compact summary and the estimated parameters.
219 print(results.short_summary())
220 pandas_results = get_pandas_estimated_parameters(
221     estimation_results=results ,
222     group_of_parameters=built_model.parameter_groups ,
223 )
224 for group_name, pandas_table in pandas_results.items():
225     print(group_name if group_name else 'Estimated parameters')
226     print(pandas_table)
227
228 general_statistics = get_latex_general_statistics(estimation_results=results)
229 print(general_statistics)
230
231 estimated_parameters = get_latex_estimated_parameters(
232     estimation_results=results ,
233     group_of_parameters=built_model.parameter_groups ,
234 )
235 for group_name, latex_table in estimated_parameters.items():
236     print(group_name if group_name else 'Estimated parameters')
237     print(latex_table)

```

The file `plot_h04_mode_lv_gauss_simult.py` implements the first simultaneous hybrid choice model of the tutorial. It combines the mode choice model with the Gaussian MIMIC measurement model in a single likelihood function. The script first defines Gaussian measurement equations for all Likert indicators, using positive scale parameters for the measurement errors. It then loads the Optima data and creates the build context used to translate the semantic latent-variable specification into Biogeme expressions.

A central part of the file is the normalization plan. As in the MIMIC model, the indicator `Envir01` is used as reference indicator: its intercept is fixed to zero, and its loading on the car-centric latent variable is fixed to -1 . These two constraints fix the location, scale, and orientation of the latent variable. The semantic model is then resolved and converted into Biogeme expressions. The utility functions are generated with the latent variable included in

the car utility, and the logit choice probability is constructed.

The main difference with the sequential specification is the construction of the likelihood. The conditional likelihood of the measurement model and the conditional likelihood of the mode choice model are multiplied conditionally on the latent variable. This combined likelihood is then integrated over the latent-variable distribution using Monte-Carlo simulation. The resulting expression is the log-likelihood of the simultaneous hybrid choice model. The last part of the script estimates the model, or reloads previously saved results, and prints the estimated parameters grouped according to their role in the model.

B.10 plot_h05_mode_lv_ordlogit_simult.py

```

1  """
2  Ordered-logit hybrid mode choice model: simultaneous maximum likelihood estimation
3  =====
4
5  This example estimates a hybrid mode choice model with one latent variable and
6  ordered-logit measurement equations. The observed Likert indicators are modeled
7  as ordinal responses, and the associated threshold parameters are estimated
8  jointly with the latent-variable and choice-model parameters.
9
10 Compared with the previous Gaussian specification, only the measurement model
11 changes: the hybrid structure, the latent variable, and the simultaneous
12 maximum-likelihood estimation strategy remain the same.
13
14 The latent-variable structure is imported from a separate semantic specification
15 file. This script adds the ordered-logit measurement configuration, the
16 normalization constraints required for identification, the mode-choice utilities,
17 and the simultaneous maximum-likelihood estimation setup.
18
19 The script performs the following steps:
20
21 - load the latent-variable and indicator specifications,
22 - define an ordered-logit measurement configuration for all indicators,
23 - define the normalization constraints used for identification,
24 - resolve the semantic specification into an estimable model,
25 - build the Biogeme expressions from the resolved model,
26 - build the choice utilities, including the latent-variable term,
27 - combine the ordered-logit measurement and choice conditional likelihoods,
28 - integrate the combined conditional likelihood over the latent variable,
29 - estimate the hybrid model, or reload previously saved estimation results,
30 - display the estimated parameters as grouped pandas and LaTeX tables.
31
32 Michel Bierlaire
33 Mon Jun 15 2026, 19:55:30
34 """
35
36 from __future__ import annotations
37
38 from choice_latent_variables import generate_utility_functions
39 from likert_spec import likert_indicators, likert_types
40 from number_of_draws import NUMBER_OF_DRAWS
41 from one_latent_variable_spec import latent_variables
42 from optima import Choice, read_data
43
44 import biogeme.biogeme_logging as blog
45 from biogeme.biogeme import BIOGEME
46 from biogeme.expressions import MonteCarlo, log
47 from biogeme.latent_variables import (
48     BuildContext,
49     EstimationMode,
50     Fixing,
51     IndicatorMeasurementSpec,
52     MeasurementConfiguration,
53     MeasurementIntercept,
54     MeasurementLoading,
55     MeasurementModel,
56     MeasurementSigma,
57     NormalizationPlan,
58     PositiveParameterSpec,
59     PositivityMode,
60     build_biogeme_model,
61     resolve_model,
62 )
63 from biogeme.models import logit
64 from biogeme.results_processing import (
65     get_latex_estimated_parameters,
66     get_latex_general_statistics,
67     get_pandas_estimated_parameters,
68 )
69
70 logger = blog.get_screen_logger(level=blog.INFO)
71
72 DEFAULT_MEASUREMENT_SIGMA_START = 10.0
73
74
75 # %%
76 # Ordered-logit measurement configuration for all Likert indicators.
77 # -----
78 measurement_configuration = MeasurementConfiguration(
79     specifications=[
80         IndicatorMeasurementSpec(
81             indicator_name=indicator.name,
82             measurement_model=MeasurementModel.ORDERED_LOGIT,
83             measurement_sigma=PositiveParameterSpec(
84                 start=DEFAULT_MEASUREMENT_SIGMA_START
85             ),
86         )
87     ]
88     for indicator in likert_indicators
89 )

```

```

89 )
90
91 # %%
92 # Load the Optima data.
93 # -----
94 database = read_data()
95
96 # %%
97 # Define the build context for maximum likelihood estimation.
98 # -----
99 # The build context specifies how the semantic latent-variable specification is
100 # translated into estimable Biogeme expressions. We start from the default
101 # maximum-likelihood context and explicitly select the log-exp parameterization
102 # for positive parameters. The ordinal options are relevant here because the
103 # measurement equations use ordered-logit probabilities.
104 default_context = BuildContext.default(EstimationMode.MAXIMUM_LIKELIHOOD)
105 context = BuildContext(
106     estimation_mode=default_context.estimation_mode,
107     draw_type=default_context.draw_type,
108     positivity_mode=PositivityMode.LOG_EXP,
109     naming=default_context.naming,
110     ordinal_eps=default_context.ordinal_eps,
111     ordinal_enforce_order=default_context.ordinal_enforce_order,
112 )
113
114 # %%
115 # Identification constraints for the latent variable
116 # -----
117 # Two layers of normalization are needed in this ordered-logit specification.
118 #
119 # 1. Ordered-logit measurement-scale normalization
120 # The latent response underlying the ordinal indicators has an arbitrary
121 # scale. We therefore fix one measurement sigma to 1.0 in order to anchor
122 # the overall scale of the ordered-logit measurement model.
123 #
124 # 2. Latent-variable normalization by reference indicator
125 # The latent variable itself is not directly observed, so its location and
126 # orientation must also be fixed. We use indicator 'Envir01' as the
127 # reference indicator:
128 # - its intercept is fixed to 0.0, which anchors the location of the latent
129 #   variable;
130 # - its loading on 'car_centric_attitude' is fixed to -1.0, which anchors
131 #   the scale and fixes the orientation of the latent variable.
132 #
133 # The negative sign means that higher values of 'car_centric_attitude' imply
134 # lower expected values for 'Envir01'. With these constraints in place, the
135 # remaining intercepts, loadings, and thresholds are interpreted relative to
136 # the chosen reference and measurement scale.
137 normalization_plan = NormalizationPlan()
138
139 normalization_plan.add(
140     Fixing(
141         MeasurementSigma('Envir01'),
142         1.0,
143         note='ordered-logit measurement model: scale normalization',
144     )
145 )
146
147 normalization_plan.add(
148     Fixing(
149         MeasurementIntercept('Envir01'),
150         0.0,
151         note='car_centric_attitude reference indicator: location',
152     )
153 )
154 normalization_plan.add(
155     Fixing(
156         MeasurementLoading('car_centric_attitude', 'Envir01'),
157         -1.0,
158         note='car_centric_attitude reference indicator: scale and orientation',
159     )
160 )
161
162 # %%
163 # Resolve the semantic specification.
164 # -----
165 # The resolver combines the latent-variable specification, the indicator
166 # definitions, the ordered-logit measurement configuration, and the normalization
167 # plan into an internal resolved model.
168 resolved_model = resolve_model(
169     latent_variables=latent_variables,
170     likert_indicators=likert_indicators,
171     likert_types=likert_types,
172     measurement_configuration=measurement_configuration,
173     context=context,
174     normalization_plan=normalization_plan,
175 )
176
177 # %%
178 # Build the Biogeme expressions.
179 # -----
180 # The builder translates the resolved model into expressions that can be used by
181 # Biogeme for simultaneous maximum-likelihood estimation. It also provides
182 # report-ready parameter groups based on the parameters that are actually
183 # estimated.
184 built_model = build_biogeme_model(resolved_model)
185
186 # %%
187 # Choice utilities including the latent-variable term.
188 # -----
189 utilities = generate_utility_functions(built_model.latent_expressions)
190
191 # %%
192 # Conditional likelihood of the mode choice model
193 # -----
194 conditional_choice_likelihood = logit(utilities, None, Choice)
195

```

```

196 # %%
197 # Combined conditional likelihood
198 #
199 # The ordered-logit measurement component and the mode-choice component are
200 # combined conditionally on the latent variable. As in the previous step,
201 # both parts are estimated jointly in a single likelihood.
202 combined_conditional_likelihood = (
203     built_model.conditional_likelihood * conditional_choice_likelihood
204 )
205
206 # %%
207 # Log-likelihood
208 #
209 # The combined conditional likelihood is integrated over the latent variable
210 # by Monte Carlo. Taking the logarithm of the resulting integrated likelihood
211 # yields the log-likelihood used for simultaneous maximum-likelihood
212 # estimation.
213 integrated_likelihood = MonteCarlo(combined_conditional_likelihood)
214 log_likelihood = log(integrated_likelihood)
215
216 # %%
217 # Estimate the model with Biogeme.
218 #
219 # Existing results are reloaded from the YAML file when available.
220 biogeme = BIOGEME(
221     database,
222     log_likelihood,
223     number_of_draws=NUMBER_OF_DRAWS,
224     calculating_second_derivatives='never',
225     max_iterations=5_000,
226     group_of_parameters=built_model.parameter_groups,
227 )
228 biogeme.model_name = 'plot_h05_mode_lv_ordlogit_simult'
229
230 yaml_file_name = f'saved_results/{biogeme.model_name}.yaml'
231 results = biogeme.estimate_or_load(yaml_file_name=yaml_file_name)
232
233 # %%
234 # Display a compact summary and the estimated parameters.
235 print(results.short_summary())
236 pandas_results = get_pandas_estimated_parameters(
237     estimation_results=results,
238     group_of_parameters=built_model.parameter_groups,
239 )
240
241 for group_name, pandas_table in pandas_results.items():
242     print(group_name if group_name else 'Estimated parameters')
243     print(pandas_table)
244
245 general_statistics = get_latex_general_statistics(estimation_results=results)
246 print(general_statistics)
247
248 estimated_parameters = get_latex_estimated_parameters(
249     estimation_results=results,
250     group_of_parameters=built_model.parameter_groups,
251 )
252 for group_name, latex_table in estimated_parameters.items():
253     print(group_name if group_name else 'Estimated parameters')
254     print(latex_table)

```

The file `plot_h05_mode_lv_ordlogit_simult.py` implements the simultaneous hybrid choice model with ordered-logit measurement equations. It starts from the same semantic specification of the car-centric latent variable used in the previous examples, and only changes the way in which the Likert indicators are modeled. Instead of treating them as continuous responses, the script declares an ordered-logit measurement model for each indicator through a `MeasurementConfiguration`.

The code is organized according to the theoretical decomposition of the hybrid choice model. First, it loads the data and defines the estimation context, including the log-exp parameterization used to enforce positivity constraints. It then specifies the normalization plan. In the ordinal case, two types of normalization are needed: one measurement scale parameter is fixed in order to identify the scale of the ordered-logit measurement layer, and the reference indicator `Envir01` is used to fix the location, scale, and orientation of the latent variable. Its intercept is fixed to zero and its loading on the car-centric attitude is fixed to -1 .

The semantic latent-variable specification, the Likert indicators, the ordered-logit measurement configuration, and the normalization plan are then resolved into an estimable model. The resulting Biogeme expressions contain the conditional likelihood of the measurement model, the latent-variable expression, and the corresponding parameter groups used later to organize the output tables.

The mode-choice component is built separately by calling `generate_utility_functions`, which inserts the latent variable into the utility of the car alternative. The standard logit choice probability is then multiplied by the ordered-logit measurement likelihood. This product is conditional on the latent variable. The final likelihood is obtained by Monte-Carlo integration over the latent variable, followed by the logarithm of the integrated likelihood.

Finally, the script creates the `BIOGEME` object, estimates the model or reloads previously

saved results, and prints the output in several formats. The use of parameter groups makes it possible to separate the reported estimates into structural parameters, measurement parameters, threshold parameters, and choice-model parameters, consistently with the theoretical structure of the hybrid choice model.

B.11 plot_h06_mode_lv_ordprobit_simult.py

```

1  """
2  Ordered-probit hybrid mode choice model: simultaneous maximum likelihood estimation
3  =====
4
5  This example estimates a hybrid mode choice model with one latent variable and
6  ordered-probit measurement equations. The observed Likert indicators are treated
7  as ordinal responses. Their thresholds, the latent-variable model, and the
8  choice-model parameters are estimated jointly.
9
10 The model is the ordered-probit counterpart of
11 'plot_h05_mode_lv_ordlogit_simult'. The hybrid structure, the latent variable,
12 the choice utilities, and the simultaneous maximum-likelihood estimation
13 strategy are the same. Only the ordinal measurement distribution changes: the
14 indicator probabilities are based on normal cumulative distributions instead of
15 logistic cumulative distributions.
16
17 The latent-variable structure is imported from a separate semantic specification
18 file. This script adds the ordered-probit measurement configuration, the
19 normalization constraints required for identification, the mode-choice utilities,
20 and the simultaneous maximum-likelihood estimation setup.
21
22 The script performs the following steps:
23
24 - load the latent-variable and indicator specifications,
25 - define an ordered-probit measurement configuration for all indicators,
26 - define the normalization constraints used for identification,
27 - resolve the semantic specification into an estimable model,
28 - build the Biogeme expressions from the resolved model,
29 - build the choice utilities, including the latent-variable term,
30 - combine the ordered-probit measurement and choice conditional likelihoods,
31 - integrate the combined conditional likelihood over the latent variable,
32 - estimate the hybrid model, or reload previously saved estimation results,
33 - display the estimated parameters as grouped pandas and LaTeX tables.
34
35 Michel Bierlaire
36 Sat Jun 13 2026, 15:13:40
37 """
38
39 from __future__ import annotations
40
41 import sys
42
43 from choice_latent_variables import generate_utility_functions
44 from likert_spec import likert_indicators, likert_types
45 from number_of_draws import NUMBER_OF_DRAWS
46 from one_latent_variable_spec import latent_variables
47 from optima import Choice, read_data
48
49 import biogeme.biogeme_logging as blog
50 from biogeme.biogeme import BIOGEME
51 from biogeme.expressions import MonteCarlo, log
52 from biogeme.latent_variables import (
53     BuildContext,
54     EstimationMode,
55     Fixing,
56     IndicatorMeasurementSpec,
57     MeasurementConfiguration,
58     MeasurementIntercept,
59     MeasurementLoading,
60     MeasurementModel,
61     MeasurementSigma,
62     NormalizationPlan,
63     PositiveParameterSpec,
64     PositivityMode,
65     build_biogeme_model,
66     generate_python_code,
67     resolve_model,
68 )
69 from biogeme.models import logit
70 from biogeme.results_processing import (
71     get_latex_estimated_parameters,
72     get_latex_general_statistics,
73     get_pandas_estimated_parameters,
74 )
75
76 logger = blog.get_screen_logger(level=blog.INFO)
77
78 DEFAULT_MEASUREMENT_SIGMA_START = 10.0
79
80 # %%
81 # Ordered-probit measurement configuration for all Likert indicators.
82 #
83 # The semantic specification, the latent variable, and the mode-choice component
84 # are the same as in the ordered-logit example. Only the ordinal measurement
85 # distribution changes: ordered logit uses the logistic CDF, whereas ordered
86 # probit uses the normal CDF.
87 measurement_configuration = MeasurementConfiguration(
88     specifications=[
89         IndicatorMeasurementSpec(
90             indicator.name=indicator.name,
91             measurement_model=MeasurementModel.ORDERED_PROBIT,
92             measurement_sigma=PositiveParameterSpec(
93                 start=DEFAULT_MEASUREMENT_SIGMA_START
94             ),
95         ),
96     ]

```

```

97     for indicator in likert_indicators
98     ]
99 )
100
101 # %%
102 # Load the Optima data.
103 # -----
104 database = read.data()
105
106 # %%
107 # Define the build context for maximum likelihood estimation.
108 # -----
109 # The build context specifies how the semantic latent-variable specification is
110 # translated into estimable Biogeme expressions. We start from the default
111 # maximum-likelihood context and explicitly select the log-exp parameterization
112 # for positive parameters. The ordinal options control the construction and
113 # numerical treatment of ordered-probit thresholds.
114 default_context = BuildContext.default(EstimationMode.MAXIMUM_LIKELIHOOD)
115 context = BuildContext(
116     estimation_mode=default_context.estimation_mode,
117     draw_type=default_context.draw_type,
118     positivity_mode=PositivityMode.LOG_EXP,
119     naming=default_context.naming,
120     ordinal_eps=default_context.ordinal_eps,
121     ordinal_enforce_order=default_context.ordinal_enforce_order,
122 )
123
124 # %%
125 # Identification constraints for the latent variable
126 # -----
127 # Two layers of normalization are needed in this ordered-probit specification.
128 # They are the same as in :mod:`plot_h05_mode_lv_ordlogit_simult`.
129 #
130 # 1. Ordinal measurement-scale normalization
131 # The latent response underlying the ordinal indicators has an arbitrary
132 # scale. We therefore fix one measurement sigma to 1.0 in order to anchor
133 # the overall scale of the ordered-probit measurement model.
134 #
135 # 2. Latent-variable normalization by reference indicator
136 # The latent variable itself is not directly observed, so its location and
137 # orientation must also be fixed. We use indicator 'Envir01' as the
138 # reference indicator:
139 # - its intercept is fixed to 0.0, which anchors the location of the latent
140 #   variable;
141 # - its loading on 'car-centric-attitude' is fixed to -1.0, which anchors
142 #   the scale and fixes the orientation of the latent variable.
143 #
144 # The negative sign means that higher values of 'car-centric-attitude' imply
145 # lower expected values for 'Envir01'. With these constraints in place, the
146 # remaining intercepts, loadings, and thresholds are interpreted relative to
147 # the chosen reference and measurement scale.
148 normalization_plan = NormalizationPlan()
149
150 normalization_plan.add(
151     Fixing(
152         MeasurementSigma('Envir01'),
153         1.0,
154         note='ordered-probit measurement model: scale normalization',
155     )
156 )
157
158 normalization_plan.add(
159     Fixing(
160         MeasurementIntercept('Envir01'),
161         0.0,
162         note='car-centric-attitude reference indicator: location',
163     )
164 )
165 normalization_plan.add(
166     Fixing(
167         MeasurementLoading('car-centric-attitude', 'Envir01'),
168         -1.0,
169         note='car-centric-attitude reference indicator: scale and orientation',
170     )
171 )
172
173 # %%
174 # Resolve the semantic specification.
175 # -----
176 # The resolver combines the latent-variable specification, the indicator
177 # definitions, the ordered-probit measurement configuration, and the
178 # normalization plan into an internal resolved model.
179 resolved_model = resolve_model(
180     latent_variables=latent_variables,
181     likert_indicators=likert_indicators,
182     likert_types=likert_types,
183     measurement_configuration=measurement_configuration,
184     context=context,
185     normalization_plan=normalization_plan,
186 )
187
188 # %%
189 # Build the Biogeme expressions.
190 # -----
191 # The builder translates the resolved model into expressions that can be used by
192 # Biogeme for simultaneous maximum-likelihood estimation. It also provides
193 # report-ready parameter groups based on the parameters that are actually
194 # estimated.
195 built_model = build_biogeme_model(resolved_model)
196
197 # %%
198 # Choice utilities including the latent-variable term.
199 # -----
200 utilities = generate_utility_functions(built_model.latent_expressions)
201
202 # %%
203 # Conditional likelihood of the mode choice model

```

```

204 # -----
205 conditional.choice.likelihood = logit(utilities , None, Choice)
206
207 # %%
208 # Combined conditional likelihood.
209 # -----
210 # The ordered-probit measurement component and the mode-choice component are
211 # multiplied conditionally on the latent variable. The product is later
212 # integrated over the latent-variable distribution, so all components are
213 # estimated jointly in one likelihood.
214 combined_conditional.likelihood = (
215     built_model.conditional.likelihood * conditional.choice.likelihood
216 )
217
218 # %%
219 # Log-likelihood.
220 # -----
221 # The combined conditional likelihood is integrated over the latent variable by
222 # Monte Carlo. Taking the logarithm of the resulting integrated likelihood yields
223 # the log-likelihood used for simultaneous maximum-likelihood estimation.
224 integrated_likelihood = MonteCarlo(combined_conditional.likelihood)
225 log_likelihood = log(integrated_likelihood)
226
227 # %%
228 # Estimate the model with Biogeme.
229 # -----
230 # Existing results are reloaded from the YAML file when available.
231 biogeme = BIOGEME(
232     database ,
233     log_likelihood ,
234     number_of_draws=NUMBER_OF_DRAWS,
235     calculating_second_derivatives='never' ,
236     max_iterations=5_000 ,
237     group_of_parameters=built_model.parameter_groups ,
238 )
239 biogeme.model_name = 'plot_h06_mode_lv_ordprobit_simult'
240
241 yaml_file_name = f'saved_results/{biogeme.model_name}.yaml'
242 results = biogeme.estimate_or.load(yaml_file_name=yaml_file_name)
243
244 # %%
245 # Display a compact summary and the estimated parameters.
246 print(results.short_summary())
247 pandas_results = get_pandas_estimated_parameters(
248     estimation_results=results ,
249     group_of_parameters=built_model.parameter_groups ,
250 )
251 for group_name, pandas_table in pandas_results.items():
252     print(group_name if group_name else 'Estimated parameters')
253     print(pandas_table)
254
255 general_statistics = get_latex_general_statistics(estimation_results=results)
256 print(general_statistics)
257
258 estimated_parameters = get_latex_estimated_parameters(
259     estimation_results=results ,
260     group_of_parameters=built_model.parameter_groups ,
261 )
262 for group_name, latex_table in estimated_parameters.items():
263     print(group_name if group_name else 'Estimated parameters')
264     print(latex_table)

```

The script `plot_h06_mode_lv_ordprobit_simult.py` implements the ordered-probit version of the simultaneous hybrid choice model. Its structure is deliberately parallel to `plot_h05_mode_lv_ordlogit_simult.py`: the latent-variable specification, the normalization constraints, the choice utilities, and the simultaneous estimation strategy are the same. The only conceptual difference is the distribution used in the ordinal measurement model. Here, the probabilities of the Likert responses are computed with the normal cumulative distribution function, leading to an ordered-probit measurement model.

The script first imports the semantic specification of the latent variable, the Likert indicators, the number of draws used for Monte-Carlo integration, and the data preparation routine. It then defines a `MeasurementConfiguration` in which each Likert indicator is associated with an `ORDERED_PROBIT` measurement model. Positive measurement scale parameters are represented using a log-exp transformation, which ensures that the corresponding standard deviations remain strictly positive during estimation.

The normalization plan contains the identifying restrictions required by the ordered-probit hybrid model. The measurement scale is anchored by fixing the scale parameter of the reference indicator `Envir01` to one. The latent variable is normalized using the same reference indicator: its intercept is fixed to zero and its loading on the car-centric attitude is fixed to -1 . These constraints determine the location, scale, and orientation of the latent variable.

The semantic specification is then resolved into an estimable model and translated into Biogeme expressions. The choice utilities are generated from the latent-variable expressions, so that the car-centric attitude can enter the utility of the car alternative. The conditional likelihood of the ordered-probit measurement model is multiplied by the conditional logit choice

probability. This product is conditional on the latent variable and is integrated by Monte-Carlo simulation. The logarithm of this integrated likelihood is the objective function used for simultaneous maximum-likelihood estimation.

Finally, the script estimates the model, or reloads previously saved results if available. The estimated parameters are then displayed in grouped pandas and L^AT_EX tables, using the parameter groups generated from the resolved model. This organization makes explicit the correspondence between the theoretical components of the hybrid choice model—structural equation, measurement equations, thresholds, and choice model—and the blocks of parameters reported in the estimation output.

B.12 plot_h07_mode_lv_ordprobit_simult_generated.py

```

1  """Simultaneous hybrid choice model with ordered-probit indicators.
2
3
4  This example estimates a hybrid choice model where a latent variable,
5  interpreted as a car-centric attitude, influences the utilities of a
6  mode-choice model. The latent variable is described by a structural equation
7  and is measured through several attitudinal indicators using ordered-probit
8  measurement equations.
9
10 The model is estimated simultaneously: the measurement equations and the
11 choice model are combined into one likelihood function. Because the latent
12 variable is not observed, the likelihood is integrated by Monte Carlo
13 simulation.
14
15 The script is organized with ‘# %%’ markers. Each marker denotes the start
16 of a new cell when the example is later converted into a notebook. New cells are
17 introduced whenever a new concept is presented, so that the generated notebook
18 can be read progressively as a pedagogical example. The comments therefore
19 describe both the statistical role of each block and the pedagogical message of
20 the corresponding notebook cell.
21
22 Tested with Biogeme 3.3.3.
23 Michel Bierlaire
24 Sat Jun 13 2026, 15:13:40
25
26 """
27
28 from choice_latent_variables import (
29     generate_utility_functions,
30 )
31 from number_of_draws import NUMBER_OF_DRAWES
32 from optima import Choice, read_data
33
34 import biogeme.biogeme_logging as blog
35 from biogeme.biogeme import BIOGEME
36 from biogeme.expressions import (
37     Beta,
38     Draws,
39     MonteCarlo,
40     MultipleProduct,
41     MultipleSum,
42     OrderedProbit,
43     Variable,
44     exp,
45     log,
46 )
47 from biogeme.models import logit
48 from biogeme.results_processing import (
49     get_latex_estimated_parameters,
50     get_latex_general_statistics,
51     get_pandas_estimated_parameters,
52 )
53
54 logger = blog.get_screen_logger(level=blog.INFO)
55
56 # %%
57 # Load the Optima data set.
58 #
59 # The data set contains the observed mode choices, explanatory variables, and
60 # attitudinal indicators used in the hybrid choice model.
61 database = read_data()
62
63
64 # %%
65 # Model parameters
66 #
67 # The parameters are organized according to their role in the model: thresholds
68 # for the Likert-scale indicators, standard deviations of measurement errors,
69 # coefficients of the measurement equations, and coefficients of the structural
70 # equation of the latent variable.
71 likert_delta_0_log = Beta('likert_delta_0_log', -0.86, None, None, 0)
72 likert_delta_0 = exp(likert_delta_0_log)
73 likert_delta_1_log = Beta('likert_delta_1_log', -0.43, None, None, 0)
74 likert_delta_1 = exp(likert_delta_1_log)
75 measurement_Envir01_sigma = 1.0
76 measurement_Envir02_sigma_log = Beta(
77     'measurement_Envir02_sigma_log', 2.302585092994046, None, None, 0
78 )
79 measurement_Envir02_sigma = exp(measurement_Envir02_sigma_log)
80 measurement_Envir06_sigma_log = Beta(
81     'measurement_Envir06_sigma_log', 2.302585092994046, None, None, 0
82 )
83 measurement_Envir06_sigma = exp(measurement_Envir06_sigma_log)
84 measurement_LifSty07_sigma_log = Beta(
85     'measurement_LifSty07_sigma_log', 2.302585092994046, None, None, 0

```

```

86 )
87 measurement_LifSty07_sigma = exp(measurement_LifSty07_sigma_log)
88 measurement_Mobil03_sigma_log = Beta(
89     'measurement_Mobil03_sigma_log', 2.302585092994046, None, None, 0
90 )
91 measurement_Mobil03_sigma = exp(measurement_Mobil03_sigma_log)
92 measurement_Mobil05_sigma_log = Beta(
93     'measurement_Mobil05_sigma_log', 2.302585092994046, None, None, 0
94 )
95 measurement_Mobil05_sigma = exp(measurement_Mobil05_sigma_log)
96 measurement_Mobil08_sigma_log = Beta(
97     'measurement_Mobil08_sigma_log', 2.302585092994046, None, None, 0
98 )
99 measurement_Mobil08_sigma = exp(measurement_Mobil08_sigma_log)
100 measurement_Mobil09_sigma_log = Beta(
101     'measurement_Mobil09_sigma_log', 2.302585092994046, None, None, 0
102 )
103 measurement_Mobil09_sigma = exp(measurement_Mobil09_sigma_log)
104 measurement_Mobil10_sigma_log = Beta(
105     'measurement_Mobil10_sigma_log', 2.302585092994046, None, None, 0
106 )
107 measurement_Mobil10_sigma = exp(measurement_Mobil10_sigma_log)
108 measurement_coefficient_car_centric_attitude_Envir01 = -1.0
109 measurement_coefficient_car_centric_attitude_Envir02 = Beta(
110     'measurement_coefficient_car_centric_attitude_Envir02', 0.0, None, None, 0
111 )
112 measurement_coefficient_car_centric_attitude_Envir06 = Beta(
113     'measurement_coefficient_car_centric_attitude_Envir06', 0.0, None, None, 0
114 )
115 measurement_coefficient_car_centric_attitude_LifSty07 = Beta(
116     'measurement_coefficient_car_centric_attitude_LifSty07', 0.0, None, None, 0
117 )
118 measurement_coefficient_car_centric_attitude_Mobil03 = Beta(
119     'measurement_coefficient_car_centric_attitude_Mobil03', 0.0, None, None, 0
120 )
121 measurement_coefficient_car_centric_attitude_Mobil05 = Beta(
122     'measurement_coefficient_car_centric_attitude_Mobil05', 0.0, None, None, 0
123 )
124 measurement_coefficient_car_centric_attitude_Mobil08 = Beta(
125     'measurement_coefficient_car_centric_attitude_Mobil08', 0.0, None, None, 0
126 )
127 measurement_coefficient_car_centric_attitude_Mobil09 = Beta(
128     'measurement_coefficient_car_centric_attitude_Mobil09', 0.0, None, None, 0
129 )
130 measurement_coefficient_car_centric_attitude_Mobil10 = Beta(
131     'measurement_coefficient_car_centric_attitude_Mobil10', 0.0, None, None, 0
132 )
133 measurement_intercept_Envir01 = 0.0
134 measurement_intercept_Envir02 = Beta(
135     'measurement_intercept_Envir02', 0.0, None, None, 0
136 )
137 measurement_intercept_Envir06 = Beta(
138     'measurement_intercept_Envir06', 0.0, None, None, 0
139 )
140 measurement_intercept_LifSty07 = Beta(
141     'measurement_intercept_LifSty07', 0.0, None, None, 0
142 )
143 measurement_intercept_Mobil03 = Beta(
144     'measurement_intercept_Mobil03', 0.0, None, None, 0
145 )
146 measurement_intercept_Mobil05 = Beta(
147     'measurement_intercept_Mobil05', 0.0, None, None, 0
148 )
149 measurement_intercept_Mobil08 = Beta(
150     'measurement_intercept_Mobil08', 0.0, None, None, 0
151 )
152 measurement_intercept_Mobil09 = Beta(
153     'measurement_intercept_Mobil09', 0.0, None, None, 0
154 )
155 measurement_intercept_Mobil10 = Beta(
156     'measurement_intercept_Mobil10', 0.0, None, None, 0
157 )
158 struct_car_centric_attitude_car_oriented_parents = Beta(
159     'struct_car_centric_attitude_car_oriented_parents', 0.0, None, None, 0
160 )
161 struct_car_centric_attitude_high_education = Beta(
162     'struct_car_centric_attitude_high_education', 0.0, None, None, 0
163 )
164 struct_car_centric_attitude_intercept = Beta(
165     'struct_car_centric_attitude_intercept', 0.0, None, None, 0
166 )
167 struct_car_centric_attitude_low_education = Beta(
168     'struct_car_centric_attitude_low_education', 0.0, None, None, 0
169 )
170 struct_car_centric_attitude_sigma_log = Beta(
171     'struct_car_centric_attitude_sigma_log', 2.302585092994046, None, None, 0
172 )
173 struct_car_centric_attitude_sigma = exp(struct_car_centric_attitude_sigma_log)
174 struct_car_centric_attitude_top_manager = Beta(
175     'struct_car_centric_attitude_top_manager', 0.0, None, None, 0
176 )
177 struct_car_centric_attitude_used_to_go_to_school_by_car = Beta(
178     'struct_car_centric_attitude_used_to_go_to_school_by_car', 0.0, None, None, 0
179 )
180
181 # %%
182 # Structural equation for the latent variable
183 #
184 # The latent variable is represented as a normal random variable. Its mean is a
185 # linear function of observed socio-economic characteristics, and its standard
186 # deviation is estimated from the data.
187 mu_car_centric_attitude = (
188     struct_car_centric_attitude_intercept
189     + struct_car_centric_attitude_top_manager * Variable('top_manager')
190     + struct_car_centric_attitude_car_oriented_parents
191     * Variable('car_oriented_parents')
192     + struct_car_centric_attitude_high_education * Variable('high_education')

```

```

193 + struct_car_centric_attitude_low_education * Variable('low_education')
194 + struct_car_centric_attitude_used_to_go_to_school_by_car
195 * Variable('used_to_go_to_school_by_car')
196 )
197 draw_car_centric_attitude = Draws(
198   'struct_car_centric_attitude_draws', draw_type='NORMAL_MLHS_ANTI'
199 )
200 car_centric_attitude = (
201   mu_car_centric_attitude
202   + struct_car_centric_attitude_sigma * draw_car_centric_attitude
203 )
204
205 # %%
206 # Threshold system for the ordered-probit indicators
207 #
208 # The attitudinal indicators are coded on a five-point Likert scale. The
209 # ordered-probit model uses four thresholds to separate the five ordered
210 # response categories. The logarithmic parameterization of the threshold
211 # increments guarantees the required ordering.
212 # Threshold system: Likert scale
213 likert_tau_1 = -(likert_delta_0 + likert_delta_1)
214 likert_tau_2 = -likert_delta_0
215 likert_tau_3 = likert_delta_0
216 likert_tau_4 = likert_delta_0 + likert_delta_1
217
218 # %%
219 # Measurement equations and ordered-probit likelihood terms
220 #
221 # Each indicator is linked to the latent variable by a linear measurement
222 # equation. The probability of the observed ordinal answer is then computed
223 # with an ordered-probit model.
224 # Indicator: Envir01
225 mu_Envir01 = (
226   measurement_intercept_Envir01
227   + measurement_coefficient_car_centric_attitude_Envir01 * car_centric_attitude
228 )
229 y_Envir01 = Variable('Envir01')
230 term_Envir01 = OrderedProbit(
231   eta=mu_Envir01 / measurement_Envir01_sigma,
232   cutpoints=[
233     likert_tau_1 / measurement_Envir01_sigma,
234     likert_tau_2 / measurement_Envir01_sigma,
235     likert_tau_3 / measurement_Envir01_sigma,
236     likert_tau_4 / measurement_Envir01_sigma,
237   ],
238   y=y_Envir01,
239   categories=[1, 2, 3, 4, 5],
240   neutral_labels=[6, -1],
241 )
242
243 # Indicator: Envir02
244 mu_Envir02 = (
245   measurement_intercept_Envir02
246   + measurement_coefficient_car_centric_attitude_Envir02 * car_centric_attitude
247 )
248 y_Envir02 = Variable('Envir02')
249 term_Envir02 = OrderedProbit(
250   eta=mu_Envir02 / measurement_Envir02_sigma,
251   cutpoints=[
252     likert_tau_1 / measurement_Envir02_sigma,
253     likert_tau_2 / measurement_Envir02_sigma,
254     likert_tau_3 / measurement_Envir02_sigma,
255     likert_tau_4 / measurement_Envir02_sigma,
256   ],
257   y=y_Envir02,
258   categories=[1, 2, 3, 4, 5],
259   neutral_labels=[6, -1],
260 )
261
262 # Indicator: Envir06
263 mu_Envir06 = (
264   measurement_intercept_Envir06
265   + measurement_coefficient_car_centric_attitude_Envir06 * car_centric_attitude
266 )
267 y_Envir06 = Variable('Envir06')
268 term_Envir06 = OrderedProbit(
269   eta=mu_Envir06 / measurement_Envir06_sigma,
270   cutpoints=[
271     likert_tau_1 / measurement_Envir06_sigma,
272     likert_tau_2 / measurement_Envir06_sigma,
273     likert_tau_3 / measurement_Envir06_sigma,
274     likert_tau_4 / measurement_Envir06_sigma,
275   ],
276   y=y_Envir06,
277   categories=[1, 2, 3, 4, 5],
278   neutral_labels=[6, -1],
279 )
280
281 # Indicator: Mobil03
282 mu_Mobil03 = (
283   measurement_intercept_Mobil03
284   + measurement_coefficient_car_centric_attitude_Mobil03 * car_centric_attitude
285 )
286 y_Mobil03 = Variable('Mobil03')
287 term_Mobil03 = OrderedProbit(
288   eta=mu_Mobil03 / measurement_Mobil03_sigma,
289   cutpoints=[
290     likert_tau_1 / measurement_Mobil03_sigma,
291     likert_tau_2 / measurement_Mobil03_sigma,
292     likert_tau_3 / measurement_Mobil03_sigma,
293     likert_tau_4 / measurement_Mobil03_sigma,
294   ],
295   y=y_Mobil03,
296   categories=[1, 2, 3, 4, 5],
297   neutral_labels=[6, -1],
298 )
299

```

```

300 # Indicator: Mobil05
301 mu_Mobil05 = (
302     measurement_intercept_Mobil05
303     + measurement_coefficient_car_centric_attitude_Mobil05 * car_centric_attitude
304 )
305 y_Mobil05 = Variable('Mobil05')
306 term_Mobil05 = OrderedProbit(
307     eta=mu_Mobil05 / measurement_Mobil05_sigma,
308     cutpoints=[
309         likert_tau_1 / measurement_Mobil05_sigma,
310         likert_tau_2 / measurement_Mobil05_sigma,
311         likert_tau_3 / measurement_Mobil05_sigma,
312         likert_tau_4 / measurement_Mobil05_sigma,
313     ],
314     y=y_Mobil05,
315     categories=[1, 2, 3, 4, 5],
316     neutral_labels=[6, -1],
317 )
318
319 # Indicator: Mobil08
320 mu_Mobil08 = (
321     measurement_intercept_Mobil08
322     + measurement_coefficient_car_centric_attitude_Mobil08 * car_centric_attitude
323 )
324 y_Mobil08 = Variable('Mobil08')
325 term_Mobil08 = OrderedProbit(
326     eta=mu_Mobil08 / measurement_Mobil08_sigma,
327     cutpoints=[
328         likert_tau_1 / measurement_Mobil08_sigma,
329         likert_tau_2 / measurement_Mobil08_sigma,
330         likert_tau_3 / measurement_Mobil08_sigma,
331         likert_tau_4 / measurement_Mobil08_sigma,
332     ],
333     y=y_Mobil08,
334     categories=[1, 2, 3, 4, 5],
335     neutral_labels=[6, -1],
336 )
337
338 # Indicator: Mobil09
339 mu_Mobil09 = (
340     measurement_intercept_Mobil09
341     + measurement_coefficient_car_centric_attitude_Mobil09 * car_centric_attitude
342 )
343 y_Mobil09 = Variable('Mobil09')
344 term_Mobil09 = OrderedProbit(
345     eta=mu_Mobil09 / measurement_Mobil09_sigma,
346     cutpoints=[
347         likert_tau_1 / measurement_Mobil09_sigma,
348         likert_tau_2 / measurement_Mobil09_sigma,
349         likert_tau_3 / measurement_Mobil09_sigma,
350         likert_tau_4 / measurement_Mobil09_sigma,
351     ],
352     y=y_Mobil09,
353     categories=[1, 2, 3, 4, 5],
354     neutral_labels=[6, -1],
355 )
356
357 # Indicator: Mobil10
358 mu_Mobil10 = (
359     measurement_intercept_Mobil10
360     + measurement_coefficient_car_centric_attitude_Mobil10 * car_centric_attitude
361 )
362 y_Mobil10 = Variable('Mobil10')
363 term_Mobil10 = OrderedProbit(
364     eta=mu_Mobil10 / measurement_Mobil10_sigma,
365     cutpoints=[
366         likert_tau_1 / measurement_Mobil10_sigma,
367         likert_tau_2 / measurement_Mobil10_sigma,
368         likert_tau_3 / measurement_Mobil10_sigma,
369         likert_tau_4 / measurement_Mobil10_sigma,
370     ],
371     y=y_Mobil10,
372     categories=[1, 2, 3, 4, 5],
373     neutral_labels=[6, -1],
374 )
375
376 # Indicator: LifSty07
377 mu_LifSty07 = (
378     measurement_intercept_LifSty07
379     + measurement_coefficient_car_centric_attitude_LifSty07 * car_centric_attitude
380 )
381 y_LifSty07 = Variable('LifSty07')
382 term_LifSty07 = OrderedProbit(
383     eta=mu_LifSty07 / measurement_LifSty07_sigma,
384     cutpoints=[
385         likert_tau_1 / measurement_LifSty07_sigma,
386         likert_tau_2 / measurement_LifSty07_sigma,
387         likert_tau_3 / measurement_LifSty07_sigma,
388         likert_tau_4 / measurement_LifSty07_sigma,
389     ],
390     y=y_LifSty07,
391     categories=[1, 2, 3, 4, 5],
392     neutral_labels=[6, -1],
393 )
394
395 # %%
396 # Conditional measurement likelihood
397 #
398 # Conditional on a realization of the latent variable, the indicator likelihood
399 # is the product of the ordered-probit probabilities. The corresponding sum of
400 # logarithms is also constructed for later combination with the choice model.
401 conditional_measurement_likelihood = MultipleProduct(
402     [
403         term_Envir01,
404         term_Envir02,
405         term_Envir06,
406         term_LifSty07,

```

```

407         term_Mobil03,
408         term_Mobil05,
409         term_Mobil08,
410         term_Mobil09,
411         term_Mobil10,
412     ]
413 )
414 conditional_log_likelihood = MultipleSum(
415     [
416         log(term)
417         for term in [
418             term_Envir01,
419             term_Envir02,
420             term_Envir06,
421             term_LifSty07,
422             term_Mobil03,
423             term_Mobil05,
424             term_Mobil08,
425             term_Mobil09,
426             term_Mobil10,
427         ]
428     ]
429 )
430 integrated_measurement_likelihood = MonteCarlo(conditional_measurement_likelihood)
431
432
433 # %%
434 # Generate the choice utilities.
435 #
436 # The latent variable is passed to the utility specification so that the
437 # car-centric attitude can directly influence the mode-choice probabilities.
438 latent_expressions = {'car_centric_attitude': car_centric_attitude}
439 utilities = generate_utility_functions(latent_expressions)
440
441 # %%
442 # Conditional choice likelihood.
443 #
444 # Given a realization of the latent variable, the probability of the observed
445 # mode choice is computed using the logit model.
446 conditional_choice_likelihood = logit(utilities, None, Choice)
447
448 # %%
449 # Joint conditional likelihood.
450 #
451 # The measurement component and the choice component are combined conditional
452 # on the latent variable. This expression is the joint contribution of one
453 # observation before integrating out the unobserved latent variable.
454 combined_conditional_likelihood = (
455     conditional_measurement_likelihood * conditional_choice_likelihood
456 )
457
458 # %%
459 # Integrate over the latent variable.
460 #
461 # The latent variable is unobserved, so the joint conditional likelihood must
462 # be integrated over its distribution. The integral is approximated by Monte
463 # Carlo simulation, and the logarithm of the simulated probability defines the
464 # log-likelihood used for estimation.
465 integrated_likelihood = MonteCarlo(combined_conditional_likelihood)
466 log_likelihood = log(integrated_likelihood)
467
468 # %%
469 # Estimate the simultaneous model with Biogeme.
470 #
471 # The dictionary groups parameters in the output tables according to their
472 # statistical role. Biogeme then estimates the model, or reloads existing
473 # results from the YAML file when available.
474 group_of_parameters = {
475     'Structural equation': [
476         'struct_car_centric_attitude_intercept',
477         'struct_car_centric_attitude_top_manager',
478         'struct_car_centric_attitude_car_oriented_parents',
479         'struct_car_centric_attitude_high_education',
480         'struct_car_centric_attitude_low_education',
481         'struct_car_centric_attitude_used_to_go_to_school_by_car',
482         'struct_car_centric_attitude_sigma_log',
483     ],
484     'Measurement equation: Envir02': [
485         'measurement_intercept_Envir02',
486         'measurement_coefficient_car_centric_attitude_Envir02',
487         'measurement_Envir02_sigma_log',
488     ],
489     'Measurement equation: Envir06': [
490         'measurement_intercept_Envir06',
491         'measurement_coefficient_car_centric_attitude_Envir06',
492         'measurement_Envir06_sigma_log',
493     ],
494     'Measurement equation: Mobil03': [
495         'measurement_intercept_Mobil03',
496         'measurement_coefficient_car_centric_attitude_Mobil03',
497         'measurement_Mobil03_sigma_log',
498     ],
499     'Measurement equation: Mobil05': [
500         'measurement_intercept_Mobil05',
501         'measurement_coefficient_car_centric_attitude_Mobil05',
502         'measurement_Mobil05_sigma_log',
503     ],
504     'Measurement equation: Mobil08': [
505         'measurement_intercept_Mobil08',
506         'measurement_coefficient_car_centric_attitude_Mobil08',
507         'measurement_Mobil08_sigma_log',
508     ],
509     'Measurement equation: Mobil09': [
510         'measurement_intercept_Mobil09',
511         'measurement_coefficient_car_centric_attitude_Mobil09',
512         'measurement_Mobil09_sigma_log',
513     ],

```

```

514     ],
515     'Measurement equation: Mobil10': [
516         'measurement_intercept_Mobil10',
517         'measurement_coefficient_car_centric_attitude_Mobil10',
518         'measurement_Mobil10_sigma_log',
519     ],
520     'Measurement equation: LifSty07': [
521         'measurement_intercept_LifSty07',
522         'measurement_coefficient_car_centric_attitude_LifSty07',
523         'measurement_LifSty07_sigma_log',
524     ],
525     'Thresholds': ['likert_delta_0_log', 'likert_delta_1_log'],
526 }
527
528 biogeme = BIOGEME(
529     database,
530     log_likelihood,
531     number_of_draws=NUMBER_OF_DRAWS,
532     calculating_second_derivatives='never',
533     max_iterations=5_000,
534     group_of_parameters=group_of_parameters,
535 )
536 biogeme.model_name = 'plot_h07_mode_lv_ordprobit_simult_generated'
537
538 yaml_file_name = f'saved_results/{biogeme.model_name}.yaml'
539 results = biogeme.estimate_or_load(yaml_file_name=yaml_file_name)
540
541 # %%
542 # Report the estimation results.
543 #
544 # The compact summary gives the main diagnostics. The estimated parameters and
545 # general statistics are also exported as pandas and LaTeX tables, which is
546 # useful when preparing reports, slides, or scientific documents.
547 print(results.short_summary())
548 pandas_results = get_pandas_estimated_parameters(
549     estimation_results=results,
550     group_of_parameters=group_of_parameters,
551 )
552 for group_name, pandas_table in pandas_results.items():
553     print(group_name if group_name else 'Estimated parameters')
554     print(pandas_table)
555
556 general_statistics = get_latex_general_statistics(estimation_results=results)
557 print(general_statistics)
558
559 estimated_parameters = get_latex_estimated_parameters(
560     estimation_results=results,
561     group_of_parameters=group_of_parameters,
562 )
563 for group_name, latex_table in estimated_parameters.items():
564     print(group_name if group_name else 'Estimated parameters')
565     print(latex_table)

```

This script contains an automatically generated implementation of the simultaneous hybrid choice model with ordered-probit measurement equations. It corresponds to the same model structure as `plot_h06_mode_lv_ordprobit_simult.py`, but the code is written explicitly rather than being assembled from higher-level semantic objects. It is therefore useful for inspecting the complete Biogeme expression generated from the model specification.

The script first loads the Optima data set and declares all parameters of the model. These include the threshold parameters of the Likert scale, the scale parameters of the measurement equations, the intercepts and loadings of the indicators, and the coefficients of the structural equation of the latent variable. Positive quantities, such as standard deviations and threshold increments, are parameterized in logarithmic form and exponentiated in the model expression.

The latent variable is then constructed from its structural equation. Its conditional mean is a linear function of socio-economic variables, and its random component is represented using Monte-Carlo draws. The ordered-probit measurement model is built indicator by indicator. For each Likert item, the script defines the latent response, the observed indicator, and the ordered-probit probability associated with the observed response. The four thresholds are generated from two positive increments, ensuring the required ordering of the response categories.

Conditional on a realization of the latent variable, the measurement likelihood is obtained as the product of the ordered-probit probabilities. The same latent variable is then inserted into the mode choice utilities through the function `generate_utility_functions`. The conditional choice probability is computed using the logit model. The measurement and choice components are multiplied to form the joint conditional likelihood, which is then integrated over the latent variable by Monte-Carlo simulation.

The last part of the script estimates the model with Biogeme. The parameters are grouped according to their statistical role, so that the output separates the structural equation, the measurement equations, the threshold parameters, and the choice-model parameters. The script finally prints the estimation summary and exports the estimated parameters both as

pandas tables and as L^AT_EX tables.

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